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An improved discrete equivalent circuit model for power assessment of bifacial half-cell photovoltaic module under front shading

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ABSTRACT

Bifacial photovoltaic modules may operate under partial shading conditions, making the electrical characteristics assessment modeling particularly important. This study aimed to develop and validate an improved discrete equivalent circuit model for bifacial half-cell photovoltaic modules that explicitly accounts for non-uniform irradiance and temperature at half-cell element level. The detailed topological circuit of bifacial half-cell photovoltaic module was implemented in MATLAB/Simulink and bidirectionally coupled with the thermal field of each half-cell element obtained from finite-volume model, enabling dynamic thermal-electrical coupling simulation under front shading conditions. The model validation was conducted through outdoor experiments under different front shading ratios and patterns, and comparisons with conventional discrete equivalent circuit and average temperature models. The results show that average temperature and conventional discrete equivalent circuit models are more suitable for predicting electrical power under non-shading conditions and cannot accurately reflect the circuit reconstruction and nonlinear current-voltage features caused by partial shading. However, improved discrete equivalent circuit model accurately restores non-uniform irradiance and temperature distributions, captures the activation of bypass diodes and the associated multi-knees current-voltage behavior. The average relative error of power and root mean square error of current are less than 6.5% and 0.56 A, respectively, for different operating conditions. This study provides a half-cell element-based modeling approach for evaluating the electrical performance of bifacial half-cell photovoltaic module, offering guidance for module layout design, shaded operation, and accurate power output prediction of bifacial photovoltaic systems.

1. Introduction

In recent years, significant advancements have been made in photovoltaic (PV) technologies, leading to an increasingly diverse range of products. Currently, PV modules are primarily categorized into monofacial photovoltaic (MPV) module and bifacial photovoltaic (BPV) module. The latter can utilize solar radiation from both front and rear sides, enabling higher energy yields and lower levelized cost of solar energy (LCOE) [1]. Meanwhile, based on the structural configuration, PV modules can be classified into full-cell elements (FCE) and half-cell elements (HCE). HCE is formed by dividing FCE into two parts, thereby reducing internal resistance and significantly enhancing module's tolerance to partial shading. This structural optimization not only improves the conversion efficiency and output power of PV modules but also enhances thermal management and reduces energy losses caused by current mismatch [2]. The adoption of HCEs can increase the output power of PV modules by up to 4.7% [3]. Moreover, HCE-based PV modules exhibit a lower LCOE [4]. It is projected that by 2030, HCE-based PV modules will occupy approximately 60% of the global market share [5].

However, PV modules still face various shading challenges during actual operation, resulting in an average annual power generation loss of 10% to 20% [6]. Partial shading can be caused by trees, nearby buildings, cloud cover, bird droppings, soil accumulation, and dust pollution. These shading factors lead to non-uniform irradiance (NUIR) and non-uniform temperature (NUT) distributions on PV modules, which in turn cause operating point deviations, current mismatch, and reduced output power [7]. In severe cases, it will even trigger hotspot

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effects [8], compromising the long-term operational stability and energy conversion efficiency of the system. Literature indicates that shading only 50% of a single cell-element can reduce the instantaneous electrical power of MPV module by more than 30% [9]. Therefore, it is important to study the electrical performance of PV modules under shading conditions to improve system reliability, optimize module selection, and improve overall system design.

For the electrical performance evaluation of PV modules, three primary modeling approaches currently exist: average-temperature (AVT) model, characteristic point model, and equivalent circuit (EC) model.

AVT model is used to estimate the instantaneous electrical efficiency of a PV module under varying temperatures based on its electrical efficiency under standard test conditions (STC) and corresponding temperature coefficient, and to further calculate the instantaneous power output. Average irradiance and average temperature of PV module are used as input parameters in this model under non-shading conditions, offering high computational efficiency and accuracy. However, it fails to capture the effects of NUIR and NUT on electrical performance. The characteristic point model estimates the instantaneous power output using two key parameters: short circuit current and open circuit voltage [10]. However, this method is established under specific conditions, limiting its adaptability and thus restricts its wide application. EC model simulates electrical performance by constructing an equivalent circuit of PV module, typically using a single-diode model [11] or a double-diode model [12]. The single-diode model, which is slightly less accurate than double-diode model, is more widely used due to its simpler formulation [13]. Compared to AVT and characteristic point models, EC model can more accurately describe the current-voltage (I-V) characteristics of PV modules under varying irradiance and temperature conditions. However, it often faces a trade-off between model accuracy and computational complexity in practical applications. Additionally, EC model either treats the entire module as a single equivalent element or accounts for spatial non-uniformity only to a limited extent, exhibiting significant limitations under complex shading scenarios and severe mismatch conditions.

Compared to traditional monofacial full-cell photovoltaic (MFCPV) modules, bifacial half-cell photovoltaic (BHCPV) modules exhibit greater complexity in circuit topology and light reception conditions. On one hand, both front and rear sides simultaneously receive multi-source irradiation including sky-diffuse radiation, direct solar radiation, and ground reflection. The spatiotemporal distribution of rear side irradiation is highly dependent on module height, ground reflectance, and surrounding environment. On the other hand, the HCE's series-parallel-series structure and bypass diodes jointly determine the current limiting, reverse bias, and circuit reconfiguration processes of each string under partial shading. This results in complex current-voltage/power-voltage (I-V/P-V) curves, exhibiting multiple characteristic knees and peaks. Existing research on BPV modules predominantly focuses on non-shading conditions and ignores the effects of irradiance and temperature non-uniformity. Systematic studies on the optical-thermal-electrical-environmental coupling behavior of BHCPV modules under typical scenarios—where the front side is partially shaded while the rear side has the inherent property of non-uniform irradiation—remain scarce. This gap is particularly evident in the absence of comprehensive modeling and experimental validation that integrates detailed circuit topologies with HCE-level NUIR/NUT distributions.

The main innovations and research methods of this study:

To address the limitations of existing studies, this study proposes an improved discrete equivalent circuit (IM-DEC) model for electrical characteristics assessment of BHCPV modules under front shading and rear non-uniform irradiation. This model accounts for NUIR and NUT distributions on cell-elements, while incorporating a dynamic thermal-electrical coupling. Firstly, NUIR distributions on front and rear sides of BHCPV module are obtained using a combination of sky-diffuse radiation and direct solar radiation models. Then, the thermal characteristics of each cell-element are computed using FLUENT software based on finite volume method (FVM). Subsequently, the electrical topology of BHCPV module is constructed in MATLAB/Simulink platform, and the electrical behavior of each HCE is simulated to obtain instantaneous electrical power and instantaneous electrical efficiency. A data interface between FLUENT and MATLAB/Simulink is established to enable coupled iterative calculations between thermal and electrical models, allowing for accurate simulation of BHCPV electrical characteristics under partial shading conditions, including bypass diode response, circuit reconstruction, and I-V/P-V curves. Finally, outdoor experimental studies are conducted under varying front shading ratios (0%, 25%, 50%, 75%) and shading patterns (vertical and horizontal shading) to validate the accuracy and applicability of IM-DEC model, while considering the effect of rear side NUIR distribution on electrical performance. Furthermore, IM-DEC model is compared with a conventional discrete equivalent circuit (CO-DEC) model and an average temperature (AVT) model used in previous studies, both of which assume uniform irradiance and temperature conditions, to show the differences in power assessment of BHCPV modules under front shading and rear non-uniform irradiation.

The content of this study is organized as follows: Section 2 reviews the literature on optical-thermal-electrical models and experimental studies of MPV and BPV under non-shading and partial shading conditions, summarizing the strengths and weaknesses of existing methods and identifying research gaps. Section 3 introduces the physical model and circuit topology of BHCPV module, along with its mathematical models, including IM-DEC model, CO-DEC model and AVT model. Section 4 describes the numerical simulation methodology. Section 5 details the experimental platform and test conditions, presenting a comparative analysis of the prediction results with three models under different shading patterns. Finally, Section 6 presents the main conclusions, and outlines the directions for future research.

2. Literature review

Studies on the electrical performance of PV modules under shading conditions primarily include experimental investigations, numerical calculations, and simulations. Compared with experiments, numerical approaches are more challenging when analyzing the performance of PV modules under partial shading conditions. This is primarily due to complex physical phenomena such as hotspot effects, the activation of bypass diodes, and mismatch losses, all of which significantly complicate the electrical analysis. Given the limitations of AVT and characteristic point models, current methods for evaluating PV performance under shading conditions are predominantly based on EC models. EC models can be classified into two categories: holistic modeling and discrete modeling. In holistic modeling, the entire PV module is treated as a single equivalent circuit element, and the variations among individual cell-elements within the module are neglected. This method simplifies the analysis by introducing effective shading factors (e.g., shading ratio), effective irradiance, or corrected current values. In contrast, in discrete modeling, each cell-element in PV module is treated as an independent electrical node, and a detailed equivalent circuit is constructed based on actual interconnection configuration (series or parallel). This method allows each cell-element to be assigned specific values for incident irradiance, temperature, and shading conditions. The following will review the current research status of both MPV and BPV modules.

2.1. For MPV researches under partial shading

Cabrera-Tobar et al. [14] explored the electrical performance of monofacial full-cell photovoltaic (MFCPV) module and monofacial half-cell photovoltaic (MHCPV) module under different shading conditions through outdoor experiments. The results showed that the power output of MHCPV module was increased by 11.3% to 20.7% compared to MFCPV module. Qian et al. [2] constructed a thermal-electrical model

to explore performance difference between MFCPV module and MHCPV module under shading conditions. However, the dynamic interaction between electrical and thermal performance was not fully considered in the model. Hanifi et al. [15] constructed an electrical model for exploring the power loss due to dust accumulation through simulation program with integrated circuit emphasis (SPICE). The electrical performance of MFCPV module and MHCPV module with different connections under shading was compared. Sundaram et al. [13] explored the mismatch loss of MFCPV module under partial shading by building a single-diode EC model in MATLAB/Simulink. The whole MFCPV module was used as the object of study in electrical model, instead of discrete cell-elements. Zhu et al. [16] proposed a simplified mathematical model for MFCPV module to predict electrical performance under partial shading. Li et al. [17] constructed a thermal-electrical model for whole MFCPV module to predict electrical performance under shading conditions. Among them, EC model was used in electrical model and shading ratio was introduced to calculate electrical performance; while thermal model did not achieve true coupling with environment. Chepp et al. [18] constructed a discrete equivalent circuit model to explore the electrical characteristics of MFCPV modules under partial shading. The study did not establish an effective optical and thermal model and there was no thermal-electric dynamic coupling. Shrivastava et al. [7] reviewed the performance and influencing factors of MPV under partial shading, covering characteristic parameters and common models, and pointed out that NUIR and NUT distributions could weaken the electrical performance. Sai Krishna and Moger [19], Pandian et al. [20] and Murugesan et al. [21] introduced reconstruction strategies to mitigate the photovoltaic shading effect.

2.2. For BPV researches under non-shading conditions

It is worth stating that since BPV technology has only been rapidly developed in recent years, compared with MPV, the research on BPV is not comprehensive enough. Most of the current studies focus on non-shading conditions, while research under partial shading conditions is relatively limited.

Shahverdian et al. [22] calculated the electrical performance of a bifacial full-cell photovoltaic (BFCPV) module using AVT model. The calculation was based on average temperature and uniform irradiation distribution, without considering the electrical characteristics inside cell-element. To simplify the computational complexity, Ewe et al. [23] and Ooshaksaraei et al. [24] used AVT model to investigated electrical performance of BFCPV modules. In these studies, the reference electrical efficiency was assumed to be identical for both front and rear sides. Gu et al. [25] used EC model to calculate electrical performance of BFCPV module, ignoring the effects of NUIR and NUT distributions on electrical properties.

2.3. For BPV researches under partial shading conditions

Some exploratory studies have been conducted on the performance of BPV modules under partial shading conditions. Ganesan et al. [26] investigated the effects of the ground distance of BFCPV module on the rear side self-shading effect, irradiance distribution and electrical performance through experiments. Polo et al. [27] applied opaque regular shading object to each stripe of commercial BPV module and replaced the bypass diode with a shunt resistor to measure the current of each stripe, thus achieving synchronous measurement of the front, rear and total irradiances; and further combined the single diode equivalent circuit model for power calculation. Hariharasudhan et al. [28] explored electrical performance of BFCPV and MFCPV modules under different shading conditions through experiments and simulations. The results showed that under same shading conditions, the power loss of BFCPV module was 26% lower than that of MFCPV module. Garrod et al. [29] used EC model to calculate electrical performance of BFCPV module, where the module had non-shading on front side and NUIR distribution on rear side. Zhang et al. [30] explored electrical performance of BFCPV module in the presence of front side shading and NUIR distribution on rear side through experiments and simulations. The results showed that front side shading posed a greater risk to BFCPV modules. Bhang et al. [31] established an EC model for BFCPV module to calculate electrical performance under partial shading conditions. The results showed that due to the simplification of model, the relative error was 42.89% when shading ratio was 50%. Raina and Sinha [32] developed a viewing factor optical model and a single-diode EC model to investigate electrical performance of BFCPV module under different front and rear sides' shading conditions. Zhang et al. [33] investigated the effect of NUIR distribution on electrical properties of BFCPV module based on EC model. The results showed that the proposed model could not accurately simulate electrical performance when there was shading on front side.

2.4. Shortcomings of the current researches

It is worth noting that the reference lumps appear in the following summary of shortcomings, but these references in lumps share common issues, and each of them has already been cited individually before this section.

1) **Optical and thermal models are often simplified or neglected.** Many studies rely solely on simplified optical models for electrical performance analysis. In thermal modeling, on one hand, the temperature distribution of PV modules is completely ignored; on the other hand, only empirical temperature coefficients are used to adjust electrical parameters. The treatment of NUIR/NUT distributions is often rather crude, making it difficult to accurately reflect actual operating conditions.
2) **Insufficient cell-element level modeling, and inadequate consideration of NUIR/NUT.** The majority of AVT and EC models are usually based on the whole module as an element, and lack discrete modeling at cell-element level. Most of these models assume uniform distributions of irradiance or temperature [13,22,29,33], thereby neglecting the effects of NUIR or NUT on electrical performance, which limits applicability in complex environmental conditions. Although a few studies [16,18,31,32] attempted to construct discrete equivalent circuit (DEC) models, most of models ignore the impact of non-uniformity on electrical performance. This simplification overlooks the variations among cell-elements and fails to accurately capture the electrical performance under partial shading scenarios.
3) **Thermal-electrical coupling is typically unidirectional or based on module averages.** Thermal-electrical coupling in many models is based on whole PV module [22,23,25,29] or ignored [13,18,27,33], and cannot reflect the dynamic thermal-electrical feedback behavior at cell-element level. Such as Ref. [17], the coupling between thermal and electrical energy is not based on an exact calculation of the electrical efficiency on each cell-element, but on an overall average of PV module. The proportion of energy converted by PV module to electrical energy is calculated using AVT model, and the remaining energy is loaded into energy equation as a boundary condition to calculate the average temperature of PV module and distributed to each cell-element. Besides, Refs. [16,28,31,32] simplify the thermal-electrical coupling process. For example, the temperature coefficient is used to correct the electrical parameters, which realizes the unidirectional effect of temperature on electrical performance. However, it is impossible to achieve the feedback process in which the temperature rise leads to a decrease in photoelectric conversion efficiency and thus affects heat generation. The bidirectional dynamic feedback between thermal model and electrical model is not established, resulting in large errors in calculation, especially under non-uniform irradiance.
4) **There is a lack of systematic research on BHCPV modules under partial shading conditions.** Existing research has primarily focused

on MFCPV, MHCPV, or BFCPV modules, with limited attention given to BHCPV modules featuring complex series-parallel-series structures and multiple bypass diode protections. Particularly under the common real-world operating condition of "partial shading on front side and NUIR on rear side", there is a lack of comprehensive models that simultaneously account for intricate circuit topologies, NUIR/NUT distributions, and dynamic thermal-electrical coupling. Furthermore, systematic comparisons and validations against outdoor experimental data remain scarce.

3. Physical and mathematical models

3.1. Physical model

The BHCPV module of EG-445M78-HD/BF-DG produced by EGING PV company is selected as the research object. The physical model of BHCPV module is shown in Fig. 1. The detailed parameters of this module are as follows: BHCPV module has a length (l), width (w), and thickness (d) of 2168 mm, 995 mm, and 35 mm, respectively. The total number of half-cell elements (HCEs) is 156, and the width (Δw) and length (Δl) of each HCE are 79.375 mm and 158.75 mm, respectively. From top to bottom, each HCE has a front glass cover of 4.2 mm thickness, a front ethylvinylacetate (EVA) encapsulation material of 0.5 mm thickness, a crystalline silicon cell (SC) of 0.4 mm thickness, a rear EVA encapsulation material of 0.5 mm thickness, and a rear glass cover of 4.2 mm thickness. The gap between upper and lower of HCE is 1.53 mm, and the gap between left and right is 2.1 mm. The dimensions of aluminum frame are shown in Fig. 1. The physical model of BHCPV module is consistent with actual conditions. The short circuit current and open circuit voltage of BHCPV module are 10.52 A and 53.37 V, respectively; the maximum power point current and voltage are 10.13 A and 43.93 V, respectively; the electrical efficiencies of front ($\eta_{ref.f}$) and rear ($\eta_{ref.r}$) sides are 20.59% and 13.38%, respectively; the bifacial factor (φ) and temperature coefficient (δ) are 65% and 0.35%, respectively. The physical properties of BHCPV module are presented in Table 1.

The installation parameters of BHCPV module are as follows: the tilt angle (θ) between ground and BHCPV module is 24°. The BHCPV module is installed facing south, with a minimum ground distance (h_{mgd}) of 960 mm. To facilitate numerical calculation, a Cartesian coordinate system is established with point O at the lower end of BHCPV module as the origin, where Z axis along the module's width direction and X axis is parallel to the ground. The ambient wind flows along the ground, forming an angle with the negative X axis, which is defined as ambient wind direction angle (α).

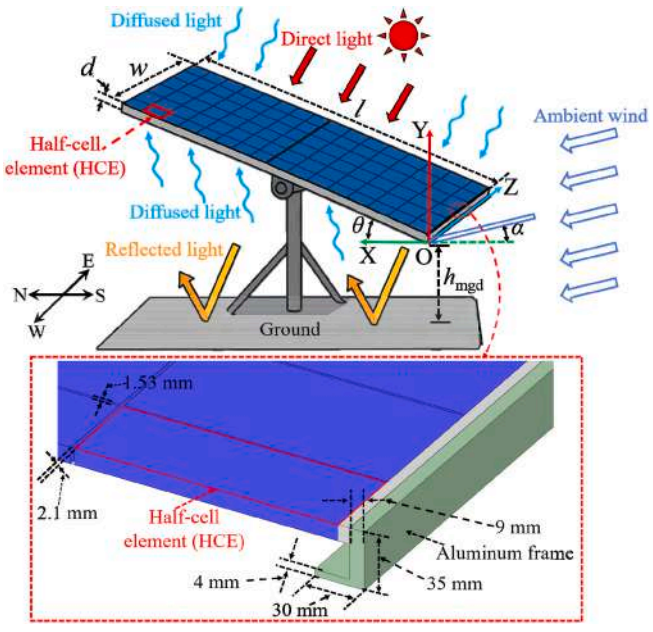


Fig. 1. Physical model of BHCPV module.

Table 1
Physical properties of BHCPV module.

Parameters	Aluminum frame	Front glass	Silicon	EVA	Rear glass
Specific heat capacity (J/kg·K)	910	850	677	2090	850
Density (kg/m^3)	2710	2530	2328	960	2530
Thermal conductivity (W/m·K)	204	0.937	150	0.35	0.937
Emissivity	0.5	0.85	-	-	0.85
Reflectivity	-	0.025	0.05	0	0.025
Transmissivity	-	0.935	0.02	1	0.915
Absorptivity	-	0.04	0.93	0	0.06

The circuit structure of BHCPV module is consistent with the electrical topology model in Refs. [2,15,34], and can be summarized as a series-parallel-series circuit. The 156 HCEs are divided into six strings, as illustrated in Fig. 2. Each string consists of 26 HCEs connected in series. Strings 1, 2, and 3 are located at upper part of module, while strings 4, 5, and 6 are at lower part. Strings 1 and 4, strings 2 and 5, and strings 3 and 6 are concatenated in parallel to form parallel groups 1, 2, and 3. These three parallel groups are then connected in series for output. Every two strings share a bypass diode: Strings 1 and 4 are connected to bypass diode 1, strings 2 and 5 to bypass diode 2, and strings 3 and 6 to bypass diode 3. This structural design helps to improve the module's ability to adapt to shading and non-uniform irradiance, and to improve output stability and reliability.

The half-cell elements and unique circuit topology in BHCPV module are key to mitigating power loss under shading, making it an ideal subject for studying local shading effects. The precision of geometric structure and circuit topology directly determines the computational accuracy of IM-DEC model. Material parameters (transmissivity, absorptivity, specific heat capacity, and density, etc.) and electrical parameters (open circuit voltage, short circuit current, and maximum power, etc.) are set based on manufacturer data and typical literature values, providing the necessary parameter basis for performance calculations. The reflectivity and absorptivity of the glass are not theoretical values for bare glass, but rather equivalent parameters considering the anti-reflective coating. These parameters are derived from average optical values measured by manufacturer under standard AM 1.5G spectrum, with an uncertainty of approximately ±2%. Additionally, the EG-445M78-HD/BF-DG module used in this study is a product launched by EGING PV company in the early stages of BPV technology development. At that stage of technological advancement, to ensure mechanical strength during lamination and installation processes, silicon wafers are commonly designed with thicker dimensions. At that time, 0.4 mm thickness is a common thickness in the industry. This value not only aligns with the manufacturing standards for early BPV modules but is also supported by relevant literature. The BPV cell thickness used in Ref. [25] is also 0.4 mm, and the glass absorptivity is 0.018, further confirming the rationality and engineering applicability of such parameters at a specific technological stage. The coordinate system is set to ensure the consistency of optical-thermal-electrical-environmental coupling model in space. These parameters and coordinate system settings constitute a tightly coupled physical basis, which is crucial for accurately capturing the optical-thermal-electrical performance under shading.

3.2. Mathematical models

3.2.1. Improved discrete equivalent circuit model (IM-DEC model)

For BHCPV modules, an accurate electrical model is a prerequisite

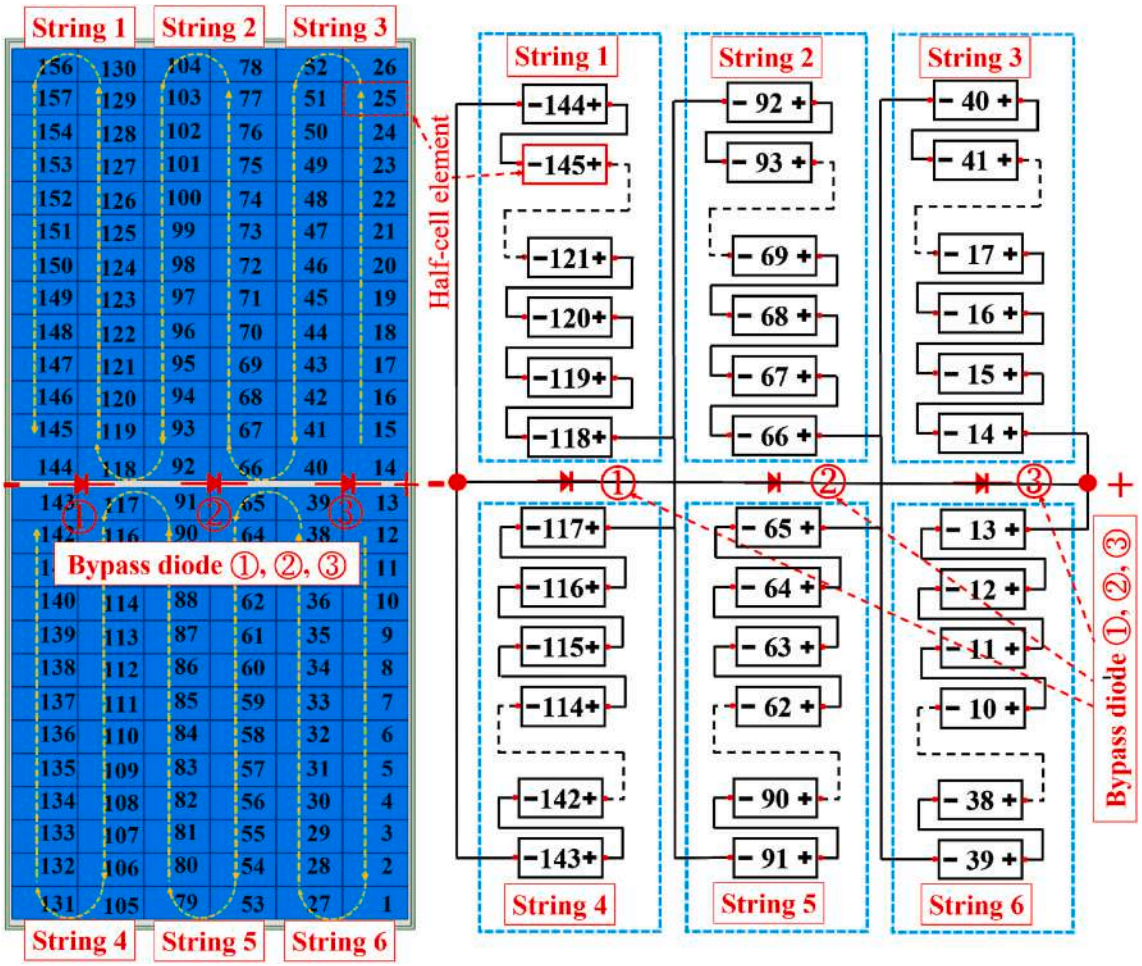


Fig. 2. Electrical topology of BHCPV module.

for predicting electrical performance. To this end, this study proposed an IM-DEC model that takes into account the non-uniformity of irradiance and temperature on HCEs and introduces a dynamic thermal-electrical coupling. IM-DEC model is compared with CO-DEC and AVT models, which assume a uniform distribution of irradiance and temperature, to explore the differences among these three models in the calculation of

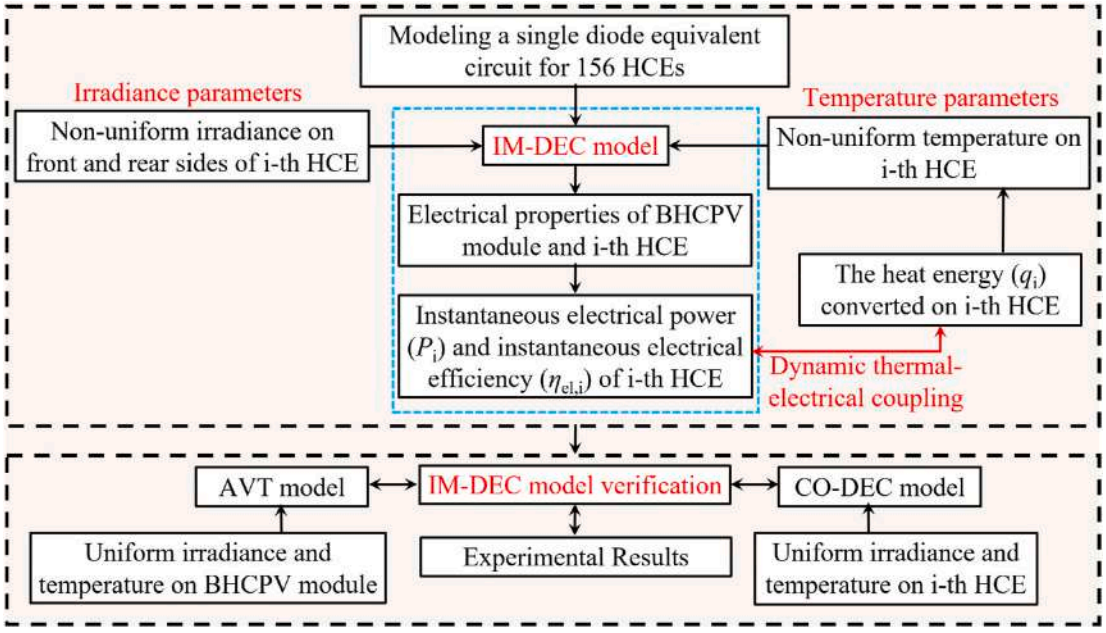


Fig. 3. Realization and validation of IM-DEC model.

electrical performance.

Due to the influence of external environment, BHCPV modules will be operated under different conditions, and the effects of irradiance and temperature differences on electrical performance of each HCE need to be considered. Therefore, IM-DEC model takes 156 HCEs as the modeling object (the total number of HCEs in BHCPV module is 156), and the specific realization and validation process is shown in Fig. 3.

Firstly, separate EC models are established for each of the 156 HCEs, and the complete electrical topology of BHCPV module is formed by connecting them in series and parallel. EC model of i-th HCE requires two key parameters to be input: equivalent irradiance value ($G_{e,i}$) and average temperature value ($t_{ave,i}$). Secondly, the complete circuit model of BHCPV module is simulated using MATLAB/Simulink platform to obtain instantaneous electric power (P_i) and instantaneous electric efficiency ($\eta_{el,i}$) of i-th HCE. Subsequently, $\eta_{el,i}$ is passed to thermal model as a boundary condition using user-defined function (UDF) of FLUENT, the proportion of solar energy converted into heat in i-th HCE is calculated, and the heat is loaded into energy equation as an internal heat source. The thermal characteristics of BHCPV module under real environments are analyzed by using FLUENT software based on FVM. Finally, the calculated $t_{ave,i}$ of i-th HCE is fed back to electrical model and used as a new input parameter to realize dynamic thermal-electrical coupling.

The above process is realized by the joint iterative operation of MATLAB/Simulink and FLUENT. The iteration is terminated after reaching the preset steady state condition to realize the accurate prediction of electrical performance. To further investigate the correctness and applicability of IM-DEC model, it will be compared with the experimental results, CO-DEC and AVT models, respectively.

3.2.2. EC model for each HCE

Before building IM-DEC model, 156 HCEs need to be modeled. As shown in Fig. 4, the single-diode EC model is a common technique for modeling the electrical properties of PV module, which consists of a current source, a diode, a series resistor (R_s), and a shunt resistor (R_{sh}). The model takes into account the electrical characteristics inside HCE, which can reflect the relationship between voltage and current in a more detailed way, as shown in Eq. (1) [35,36]:

$$I = I_{ph} - I_d - I_{sh} = I_{ph} - I_0\left[\exp\left(\frac{q(V+IR_s)}{nk(t_{ave}+273)}\right)-1\right] - \frac{V+IR_s}{R_{sh}} \tag{1}$$

where I_d and I_{sh} are partial currents flowing through diode and shunt resistor, respectively; I_{ph} is photocurrent; I_0 is diode reverse saturation current; k is Boltzmann constant (1.381×10^{-23} J/K); q is electron charge (1.602×10^{-19} C); n is diode ideal factor; and t_{ave} is average temperature of HCE in degrees Celsius. The temperature unit of the exponential term in Eq. (1) is Kelvin. Therefore, in Eq. (1), t_{ave} is converted to Kelvin temperature by (t_{ave}+273). Similarly, in the following formulas, if the temperature unit is Kelvin, the conversion is performed in the same way.

According to Ref. [37], the values of parameters in Eq. (1) are solved by following Eqs. (2)–(8) under non-standard conditions after the environmental parameters are changed. The solution for I_{ph} is shown in Eq. (2):

$$I_{ph} = \frac{G_e}{G_{ref}}\left[I_{ph,ref} + \alpha_{Isc}(t_{ave}-25)\right] \tag{2}$$

where α_{Isc} is short circuit current temperature coefficient; G_e is equivalent irradiance; G_{ref} is solar irradiance of 1000 W/m^2 under STC; $I_{ph,ref}$ is photocurrent under STC. Based on Refs. [38,39], it is known that $I_{ph,ref}$ is approximately equal to short circuit current ($I_{sc,ref}$) under STC. G_e is calculated as follows [26]:

$$G_e = G_f + G_r \times \varphi \tag{3}$$

where φ is bfaciality factor, G_f and G_r are average irradiance received on front and rear sides of HCE, respectively.

Series resistor (R_s) in Eq. (1):

$$R_s = R_{s,ref}\frac{(t_{ave}+273)}{(t_{ref}+273)}\left[1-\alpha_{Rs}\ln\left(\frac{G_e}{G_{ref}}\right)\right] \tag{4}$$

where $R_{s,ref}$ is series resistance under STC; α_{Rs} is the temperature coefficient of internal series resistance; t_{ref} is reference temperature of 25 °C.

Shunt resistor (R_{sh}) in Eq. (1):

$$R_{sh} = R_{sh,ref}\frac{G_{ref}}{G_e} \tag{5}$$

where $R_{sh,ref}$ is shunt resistor under STC.

Diode reverse saturation current (I_0) in Eq. (1):

$$I_0 = I_{0,ref}\left[\frac{(t_{ave}+273)}{(t_{ref}+273)}\right]^3 \exp\left\{\frac{q}{k}\left[\frac{E_{g,ref}}{(t_{ref}+273)} - \frac{E_g}{(t_{ave}+273)}\right]\right\} \tag{6}$$

where E_g is material energy band gap, and $I_{0,ref}$ is diode reverse saturation current under STC. The solutions of E_g and $I_{0,ref}$ are shown in Eq. (7) and Eq. (8), respectively.

$$E_g = E_{g,ref}\left[1 - 0.0002677\left(t_{ave} - t_{ref}\right)\right] \tag{7}$$

where $E_{g,ref}$ is material energy band gap of 1.121 eV at reference temperature (25 °C) [37].

Diode reverse saturation current ($I_{0,ref}$) [38,39] in Eq. (6):

$$I_{0,ref} = \frac{I_{sc,ref} - V_{oc,ref}/R_{sh,ref}}{\exp\left[\frac{qV_{oc,ref}}{nk\left(t_{ref}+273\right)}\right]-1} \tag{8}$$

where $V_{oc,ref}$ is open circuit voltage under STC.

Based on HCE equivalent circuit model established above, a complete IM-DEC model is further constructed. Meanwhile, CO-DEC and AVT models are also constructed to explore the differences between IM-DEC model and existing models under different shading ratios and shading patterns.

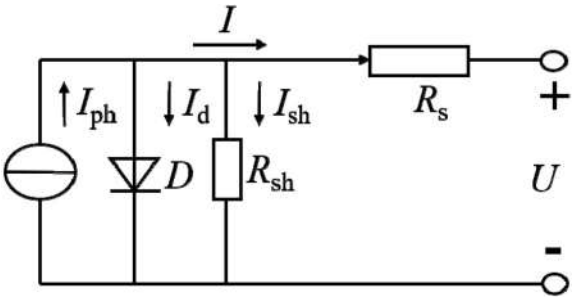


Fig. 4. Equivalent circuit model for HCE.

3.2.3. IM-DEC model

An EC model is built for each HCE shown in Fig. 2. By connecting these HCE-level EC models in series and in parallel, IM-DEC model of BHCPV module is constructed. In IM-DEC model, BHCPV module has a series-parallel-series topology: three parallel groups are connected in series for output, and each parallel group consists of two series-connected strings connected in parallel. According to Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL), the current in a series string is equal, and the voltage is the sum of the voltages of each HCE; the voltage in a parallel group is equal, and the current is the sum of the currents in each string. Taking string 1 as an example, its output current (I_1) and voltage (V_1) can be described by the following Eq. (9). The current and voltage of other strings can also be calculated by the same method.

When string 1 works normally, since the 26 HCEs in string 1 are connected in series, the current is I_1, while the voltage (V_1) of string 1 is the sum of the voltages across all 26 series HCEs [34].

$$I_1 = I_{\mathrm{HCE},118} = I_{\mathrm{HCE},119} = \ldots = I_{\mathrm{HCE},145} = I_{\mathrm{HCE},144},\\ V_1 = V_{\mathrm{HCE},118} + V_{\mathrm{HCE},119} + \ldots + V_{\mathrm{HCE},145} + V_{\mathrm{HCE},144} \tag{9}$$

When strings 1 and 4 both work normally, the current ($I_{1,4}$) and voltage ($V_{1,4}$) of strings 1 and 4 connected in parallel are as follows [34]:

$$I_{1,4} = I_1 + I_4, V_{1,4} = V_1 = V_4 \tag{10}$$

where I_1 and I_4, V_1 and V_4 are the current and voltage of strings 1 and 4, respectively.

The current (I_{total}), voltage (V_{total}), and maximum instantaneous power (P) of BHCPV module are as follows [34]:

$$I_{\mathrm{total}} = I_{1,4} = I_{2,5} = I_{3,6}, V_{\mathrm{total}} = V_{1,4} + V_{2,5} + V_{3,6}, P = I_{\mathrm{total}} \times V_{\mathrm{total}} \tag{11}$$

where $I_{2,5}$ and $I_{3,6}$, $V_{2,5}$ and $V_{3,6}$ are the current and voltage of strings 2 and 5, and strings 3 and 6 connected in parallel, respectively.

Under partial shading conditions, when the reverse bias voltage of the parallel group (such as strings 1 and 4) exceeds the conduction threshold of bypass diode (usually −0.7 V), the bypass diode 1 is triggered and becomes conductive. From the module terminal point of view, this causes the parallel group to be electrically equivalent to a conducting diode, with its voltage clamped around V_{D}. The corresponding current and voltage of BHCPV module are given by Eq. (12) [34]:

$$I_{\mathrm{total}} = I_{\mathrm{D}} = I_{2,5} = I_{3,6}, V_{\mathrm{total}} = V_{\mathrm{D}} + V_{2,5} + V_{3,6} \tag{12}$$

where I_{D} and V_{D} are the current and voltage of bypass diode.

At circuit level, the internal currents of partially shaded strings are still solved from their single-diode equations, while the series-loop current is mainly carried by the bypass diode and consistently coupled with the other two parallel groups through Eq. (12), rather than artificially forcing the substring current to zero. In numerical solution process, the module voltage is scanned point by point. At each operating point, the equations of the string single diode, parallel group, and bypass branch are combined to obtain the corresponding module terminal current. When the reverse voltage of a certain parallel group reaches the bypass threshold, the circuit automatically switches to the topology described by Eq. (12). Therefore, it can accurately reflect the redistribution of current paths in the actual circuit, thus realistically reproducing the typical current steps and characteristic knees on I-V curve caused by partial shading. To prove that this rule maintains the validity of KCL/KVL during bypass diode switching and can reproduce I-V curve, experimental verification is carried out in Section 5.

In IM-DEC model, the equivalent irradiance of i-th HCE is $G_{\mathrm{e,i}}$:

$$G_{\mathrm{e,i}} = G_{\mathrm{f,i}} \times \left(1 - \xi_{\mathrm{f,i}}\right) + G_{\mathrm{r,i}}\left(1 - \xi_{\mathrm{r,i}}\right) \times \varphi \tag{13}$$

$$\zeta_{\mathrm{f,i}} = \frac{S_{\mathrm{f,i}}}{\Delta l \Delta w}, \zeta_{\mathrm{r,i}} = \frac{S_{\mathrm{r,i}}}{\Delta l \Delta w} \tag{14}$$

where $G_{\mathrm{f,i}}$, $\zeta_{\mathrm{f,i}}$ and $S_{\mathrm{f,i}}$ are average irradiance received, shading ratios and shading areas on front side of i-th HCE, respectively; $G_{\mathrm{r,i}}$, $\zeta_{\mathrm{r,i}}$ and $S_{\mathrm{r,i}}$ are average irradiance received, shading ratios and shading areas on rear side of i-th HCE, respectively; Δl and Δw are length and width of HCE, respectively.

The instantaneous electrical efficiency ($\eta_{\mathrm{el,i}}$) of i-th HCE:

$$\eta_{\mathrm{el,i}} = \frac{P_{\mathrm{i}}}{\left[G_{\mathrm{f,i}}\left(1 - \zeta_{\mathrm{f,i}}\right) + G_{\mathrm{r,i}}\left(1 - \zeta_{\mathrm{r,i}}\right)\right]\Delta l \Delta w} \tag{15}$$

When BHCPV module reaches the maximum instantaneous electric power (P), the instantaneous electric power of i-th HCE is P_{i}. Note that HCE is not necessarily at the maximum instantaneous power when BHCPV module reaches the maximum instantaneous power.

The energy converted into heat energy in i-th HCE is q_{i}. The calculations take into account the absorption loss of the front and rear sides' encapsulant/glass, but do not consider the reflection/thermalization beyond transmittance and absorptivity. Wavelength-dependent losses are also neglected. It should be noted that this study mainly focuses on regular shading, and the bypass diode can respond well. Therefore, the heat generated by reverse bias in internal circuit is small and can be ignored. Ref. [16] also ignored this heating effect when studying regular shading.

$$q_{\mathrm{i}} = \left(1 - \eta_{\mathrm{el,i}}\right)\left[G_{\mathrm{f,i}}\left(1 - \zeta_{\mathrm{f,i}}\right)\tau_{\mathrm{g,f}}\beta + G_{\mathrm{r,i}}\left(1 - \zeta_{\mathrm{r,i}}\right)\tau_{\mathrm{g,r}}\beta\right] \tag{16}$$

where $\tau_{\mathrm{g,f}}$ and $\tau_{\mathrm{g,r}}$ are transmittance of front glass and rear glass respectively; β is absorptivity of SC layer. The optical parameters $\tau_{\mathrm{g,f}}$, $\tau_{\mathrm{g,r}}$, and β are average transmittance and absorptivity measured by manufacturer under the standard AM 1.5G solar spectrum, with an uncertainty of approximately ±2%.

3.2.4. CO-DEC model

In CO-DEC model, the equivalent irradiation of i-th HCE is $G_{\mathrm{e,i}}$:

$$G_{\mathrm{e,i}} = G_{\mathrm{BHCPV,f}} \times \left(1 - \xi_{\mathrm{f,i}}\right) + G_{\mathrm{BHCPV,r}}\left(1 - \xi_{\mathrm{r,i}}\right) \times \varphi \tag{17}$$

where $G_{\mathrm{BHCPV,f}}$ and $G_{\mathrm{BHCPV,r}}$ are average irradiance received by front and rear side of BHCPV module under non-shading, respectively.

The instantaneous electrical efficiency ($\eta_{\mathrm{el,i}}$) of i-th HCE

$$\eta_{\mathrm{el,i}} = \frac{P_{\mathrm{i}}}{\left[G_{\mathrm{BHCPV,f}}\left(1 - \zeta_{\mathrm{f,i}}\right) + G_{\mathrm{BHCPV,r}}\left(1 - \zeta_{\mathrm{r,i}}\right)\right]\Delta l \Delta w} \tag{18}$$

The energy converted into heat energy in i-th HCE is q_{i}:

$$q_{\mathrm{i}} = \left(1 - \eta_{\mathrm{el,i}}\right)\left[G_{\mathrm{BHCPV,f}}\left(1 - \zeta_{\mathrm{f,i}}\right)\tau_{\mathrm{g,f}}\beta + G_{\mathrm{BHCPV,r}}\left(1 - \zeta_{\mathrm{r,i}}\right)\tau_{\mathrm{g,r}}\beta\right] \tag{19}$$

The implementation form of CO-DEC model is similar to that of IM-DEC model. However, the primary distinction lies in the treatment of input parameters. When considering shading effects, CO-DEC model assigns a uniform equivalent irradiance to all shaded HCEs, that is, the average irradiance on front and rear sides of BPV module under non-shading conditions is multiplied by shading ratio. For non-shading HCEs, the same average irradiance is directly applied. Additionally, CO-DEC model does not consider temperature differences among individual HCEs. Instead, it uses the overall average temperature of BHCPV module, obtained from thermal model, as a uniform input for all HCEs in electrical model, rather than assigning individual average temperature of each HCE. CO-DEC model has been used in Refs. [16,18,31].

3.2.5. Average temperature electrical model (AVT model)

AVT model is widely used to calculate electrical performance of BPV modules. The model focuses on the efficiency of solar energy conversion to electrical energy from the perspective of overall energy conversion of the system. The calculation formulas for instantaneous electrical efficiency on front ($\eta_{\mathrm{el,f}}$) and rear ($\eta_{\mathrm{el,r}}$) sides of BHCPV module are as follows:

$$\eta_{\mathrm{el,f}} = \eta_{\mathrm{ref,f}}\left[1 - \delta\left(t_{\mathrm{ave}} - t_{\mathrm{ref}}\right)\right], \eta_{\mathrm{el,r}} = \eta_{\mathrm{ref,r}}\left[1 - \delta\left(t_{\mathrm{ave}} - t_{\mathrm{ref}}\right)\right] \tag{20}$$

where $\eta_{\mathrm{ref,f}}$ and $\eta_{\mathrm{ref,r}}$ are reference electrical efficiencies on front and rear sides of BHCPV module, respectively; t_{ref} is reference temperature of 25 °C; δ is temperature coefficient; t_{ave} is average temperature of BHCPV module. The maximum instantaneous electrical power (P) of BHCPV module is calculated as follows:

$$P = \left[\eta_{\mathrm{el,f}} G_{\mathrm{BHCPV,f}}\left(1 - \zeta_{\mathrm{f}}\right)\tau_{\mathrm{g,f}} + \eta_{\mathrm{el,r}} G_{\mathrm{BHCPV,r}}\left(1 - \zeta_{\mathrm{r}}\right)\tau_{\mathrm{g,r}}\right] m \Delta l \Delta w \beta \tag{21}$$

where m is the number of HCEs in BHCPV module, which is 156; ζ_{f} and ζ_{r} are shading ratios on front and rear sides of BHCPV module, respectively, which are as follows:

$$\zeta_f = \frac{S_f}{m\Delta l \Delta w}, \zeta_r = \frac{S_r}{m\Delta l \Delta w} \tag{22}$$

where S_f and S_r are shading areas on front and rear sides of BHCPV module, respectively.

The energy converted into heat energy in SC layer is q_{SC}:

$$q_{SC} = \left[(1-\eta_{el,f})G_{BHCPV,f}(1-\zeta_f)\tau_{g,f}\beta\right] + \left[(1-\eta_{el,r})G_{BHCPV,r}(1-\zeta_r)\tau_{g,r}\beta\right] \tag{23}$$

The implementation form of AVT model differs significantly from that of IM-DEC and CO-DEC models. Its core approach lies in modeling the whole BHCPV module as a single element, using average irradiance and average temperature as input parameters. The irradiance contribution is adjusted based on the shading ratio to predict the module's overall electrical performance. However, the model ignores the effect of non-uniformity on electrical performance and cannot obtain electrical output characteristics of each HCE. AVT model has been used in Refs. [22–24].

3.2.6. Acquisition of non-uniform irradiance and temperature distributions

This study focuses on proposing an IM-DEC model to predict electrical performance of BHCPV module. Therefore, optical and thermal modelings are not the focus. Instead, irradiance and temperature parameters required by IM-DEC model will be obtained using well-established models from literatures.

For optical model: The Solar Emulator module in Tracepro software based on Monte Carlo ray tracing method is used to explore the optical properties of BHCPV module. The irradiance values received by front and rear sides of BHCPV module are obtained by establishing a direct solar radiation model and a sky diffuse radiation model respectively. The sky diffuse radiation model is based on Lgawa all-sky model [40]. In direct solar radiation model, the light is set to parallel light and the irradiance is direct normal irradiance (G_{DNI}). The calculation methods of G_{DNI}, global horizontal irradiance (G_{GHI}) and diffuse horizontal irradiance (G_{DHI}) can be found in Refs. [41,42]. It has been verified that when the total number of rays reaches 44 million, further increases in the number of rays have little impact on the optical results.

The optical model takes into account the angular effects of light rays. The incident angle of light rays is precisely calculated based on the sun's position (altitude and azimuth angles) and module's installation parameters (tilt and installation azimuth angles). Simultaneously, the model considers the non-uniform distribution of sky brightness, enabling it to more realistically reflect the directional characteristics of reflected and scattered lights, and accurately capture key optical behaviors such as reflection loss and shading changes caused by variations in the incident angle.

It should be noted that the optical parameters used in Tracepro simulation are average values measured by manufacturer under the standard AM 1.5G spectrum, without considering the influence of wavelength. This trade-off is mainly to balance the complexity and computational cost of the model while ensuring that the main distribution characteristics of NUIR are revealed. Refs. [23,25] also ignored the influence of wavelength and preserved constant parameters in their studies on BPV. To further improve the accuracy of optical simulation, spectral effects can be considered in subsequent studies to reduce the underestimation of the NUIR for BPV modules.

For thermal model: To investigate the thermal characteristics of BHCPV module in a coupled environment, a sufficiently large rectangular cavity is constructed around BHCPV module to simulate the fluid domain (air), and UDF is used to achieve dynamic thermal-electrical coupling. The radiative heat transfer is calculated based on Discrete-ordinate radiation model. The velocity inlet boundary is set and the velocity vectors in X-axis and Z-axis directions are adjusted to simulate different wind velocities and directions. The shaded part on BHCPV module is considered as adiabatic, and non-shading part is set as coupling boundary. Since the Reynolds number exceeds the critical value, the Realizable κ-ε turbulence model with enhanced wall treatment is selected. The following assumptions are made before the calculation: 1) The physical properties of BHCPV module do not change with temperature [25]; 2) Previous researches have shown that interlayer reflection has a relatively small impact on PV performance. Therefore, considering the complexity and cost of calculation, interlayer reflection of BHCPV module is ignored in this study, and the same assumption is adopted in Ref. [43]; 3) The air is an incompressible ideal gas and does not participate in radiative heat transfer [44]. Detailed thermal modeling can be found in our previous study [45]. Grid independence is verified based on the average temperature of BHCPV module. When the number of grids exceeds 3,007,837, further increasing the grid number has a negligible effect on computational results.

The correctness of the optical and thermal models has been verified based on experimental results of BHCPV module under non-shading conditions on July 22, 2024 (partly cloudy, windless environment) and August 27, 2024 (sunny-dominated with few clouds, windy environment). The results show that, for both partly cloudy, windless environment and sunny-dominated with few clouds, windy environment, the maximum relative errors of irradiance distribution on front and rear sides of BHCPV module between optical model and experiments are less than 13 %; the maximum relative errors between local temperature of measurement points, and average temperature of measurement points on front and rear sides of BHCPV module, calculated using thermal model and experiments are less than 12 %. Therefore, optical and thermal model have good generalizability and credibility for performance evaluation of BHCPV systems under different environmental conditions. Detailed validations of the optical and thermal models can be found in our previous work [46]. This study will further verify the correctness of these models under shading, as shown in Sections 5.2 and 5.3.

4. Numerical solution methods

Fig. 5 shows the solution method of IM-DEC model. In first step, EC models are established for all HCEs using MATLAB/Simulink platform, and a complete BHCPV electrical topology is constructed. In second step, each EC model is provided with two input parameters: equivalent irradiance ($G_{e,i}$) and average temperature ($t_{ave,i}$). In first iteration ($j = 1$), the ambient temperature (t_{amb}) is used as the initial value for all HCEs. In third step, P_i and $\eta_{el,i}$ of each HCE are calculated based on Section 3.2.3, as well as P of BHCPV module. In fourth step, $\eta_{el,i}$ is loaded into thermal model as a boundary condition, $t_{ave,i}$ of each HCE is calculated by FLUENT, and fed back to electrical model as a new temperature parameter to start the next round of thermal-electrical coupling iterative calculation. Convergence tolerance, and stability check include two aspects: In FLUENT, the residuals of continuity and momentum equations are set to 10^{-3}, and the residuals of radiation and energy equations are set to 10^{-6}. The temperature field is considered convergent when the maximum relative change of $t_{ave,i}$ in two consecutive iterations is below the tolerance, and the criterion is shown in Eq. (24). In MATLAB/Simulink, the I-V/P-V curves are obtained by stepping the voltage. The convergence threshold for the single-point relative imbalance is set to 10^{-6}. The relative change of P and P_i need to satisfy the criterion in Eq. (25). If the convergence criterion is satisfied, the iteration is terminated and the temperature and electrical results are output. Otherwise, the simulation is continued by returning to the second step. Affected by the distributions of NUIR and UNT, the output power curve of the module will become complicated [39,47]. In each iteration, the I-V curve is obtained by numerical calculation, and the maximum instantaneous electric power (P) of BHCPV module is extracted from the corresponding P-V curve. It should be emphasized that this study is not dedicated to improving the maximum instantaneous electric power tracking efficiency of module through algorithms.

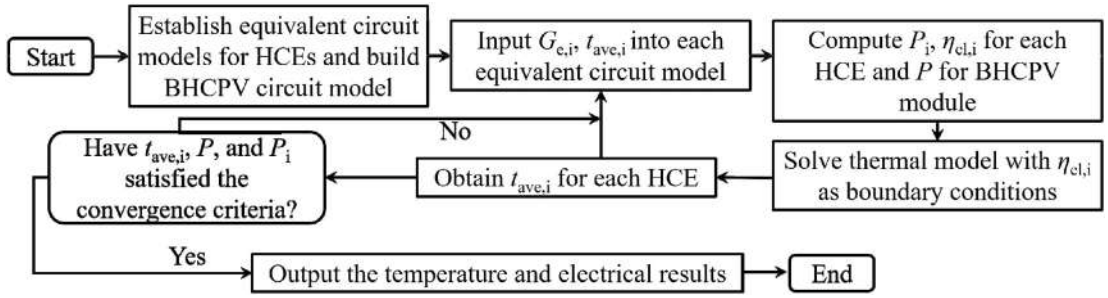


Fig. 5. Numerical solution methods of IM-DEC model.

$$\frac{\max\left|t_{ave,i}^{j}-t_{ave,i}^{j-1}\right|}{\max\left|t_{ave,i}^{j}\right|}\le 10^{-3} \tag{24}$$

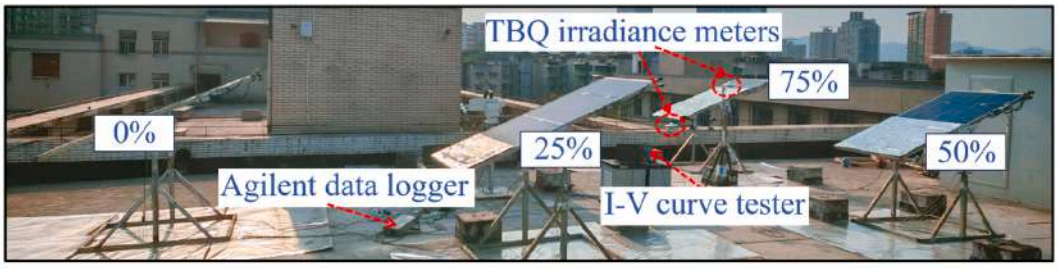
$$\frac{\max\left|P_{i}^{j}-P_{i}^{j-1}\right|}{\max\left|P_{i}^{j}\right|}\le 10^{-3},\frac{\max\left|P^{j}-P^{j-1}\right|}{\max\left|P^{j}\right|}\le 10^{-3} \tag{25}$$

In the implementation of dynamic coupling iteration, the solution processes of MATLAB/Simulink and FLUENT are controlled by main control program, respectively. Two shared databases are established to achieve data interaction: Database 1 stores average temperatures ($t_{ave,i}$) of 156 HCEs, and Database 2 stores electrical efficiencies ($\eta_{el,i}$) of HCEs. After initialization, a thermal calculation is performed first, and the temperatures of 156 HCEs are written to Database 1. Then, the thermal model is paused and waits for an update in Database 2. An update to Database 1 triggers the calculation of electrical model. MATLAB/Simulink reads the latest temperature field and writes the calculation result $\eta_{el,i}$ to Database 2, which in turn triggers FLUENT to read the new boundary conditions for next iteration. This process repeats until the convergence conditions given by Eqs. (24) and (25) are met. For operating conditions with high shading ratios and strong non-uniform influences, to avoid numerical oscillations in temperature and power during the coupling process, appropriate under-relaxation techniques are used in both thermal and electrical models, thereby improving the stability and convergence of iterative process.

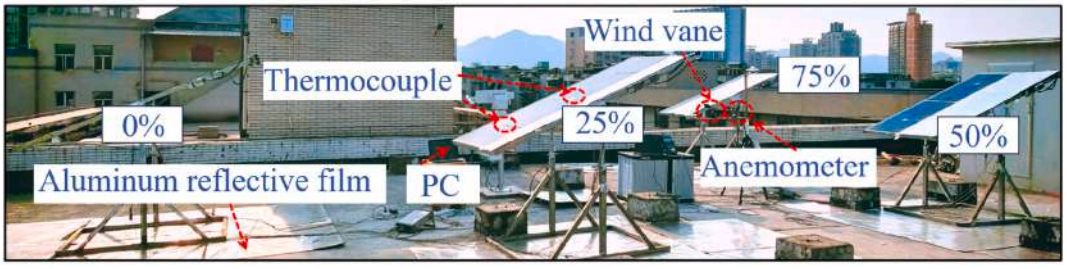
Regarding computational efficiency, on a workstation equipped with an AMD EPYC 7542 32-core processor, completing one iteration takes approximately 60 to 100 s, depending on the module's topology, operating conditions, and grid size. For example, when shading ratio is lower, completing one coupled iteration takes approximately 60-80 s, with about 300-400 iterations, resulting in a total computation time of about 5 to 8 h. When shading ratio is higher, completing one coupled iteration takes approximately 80-100 s, with about 400-500 iterations, resulting in a total computation time of about 8 to 14 h. Under higher shading ratio, the iteration time will increase accordingly due to enhanced non-uniformity.

5. Experimental testing and IM-DEC model validation

To verify the computational accuracy and applicability of IM-DEC model, and compare it with the existing CO-DEC and AVT models, this study designs and carries out outdoor shading experiments (only front shading is considered). Specifically, the electrical performance of four BHCPV modules with horizontal shading ratios (0%, 25%, 50%, 75%) on September 13, 2024, and that with vertical shading ratios (0%, 25%, 50%, 75%) on September 14, 2024 are experimentally investigated both



(a)



(b)

Fig. 6. Performance test platforms of BHCPV modules: (a) Horizontal shading; (b) Vertical shading.

from 10:00 to 16:00, respectively. All tests are conducted under clear-sky conditions. The types and parameters of four BHCPV modules are the same. The performance test platforms are shown in Fig. 6(a) and (b).

It is worth noting that shading patterns in real-world scenarios are often more complex (such as diagonal, random, or time-varying shadings), and most shading objects are not completely opaque but rather manifest as shadows that induce NUIR distributions and consequently aggravate mismatch effects. Compared to the shading patterns in practical applications, the shading scenarios considered in this study are relatively simple, and the reasons for this simplification are as follows:

Firstly, to ensure clear mechanism identification and reproducible experimental comparisons, this study selects horizontal and vertical regular shadings as standardized benchmark cases. By setting different shading ratios, typical mismatch behaviors of BHCPV module (such as reverse bias, bypass diode conduction, and circuit reconfiguration) are triggered under controlled conditions. This approach validated the applicability and computational accuracy of IM-DEC model in the chain of "non-uniform input-circuit reconfiguration-power output".

Secondly, adopting completely opaque shading objects can be regarded as a critical operating condition: if shading objects with different transmittances (i.e., shadows are generated) are introduced directly, additional uncertainties may be introduced while altering the equivalent irradiance, which may render the attribution of bypass diode conduction and thermal-electrical response mechanisms less clear; therefore, this study prioritizes opaque shading objects as the benchmark to highlight the key mismatch mechanisms.

Finally, it should be emphasized that IM-DEC model in this study is developed based on a HCE-level discretization and a dynamic thermal-electrical coupling framework. Taking $G_{e,i}$ and $t_{ave,i}$ of each HCE as input, the model can reflect the optical-thermal-electrical performance of BHCPV module under realistic NUIR/NUT distributions. Therefore, in principle, this improved model can be naturally extended to more complex shading patterns and shadow-based shading scenarios. For example, when the shading object has a certain transmittance, the front side irradiance will becomes a value between 0 and the non-shading irradiance, which can be accurately calculated based on the transmittance of shading object. Therefore, IM-DEC model can directly simulate the mismatch effects caused by shading objects with different transmittance without any structural modifications. However, the robustness and accuracy of IM-DEC model under complex shading scenarios still requires further experimental validation, which is the main focus for future work. Similarly, Refs. [16,27,34] also adopted regular opaque shadings when investigating the shading performance of photovoltaic modules.

To ensure that each BHCPV module has the same ground reflectance characteristics, an aluminum film with a reflectance of 0.9 is laid underneath each module, and the area of aluminum film is approximately three times that of BHCPV module. Figs. 7 and 8 are schematic diagrams of horizontal and vertical shadings under different shading ratios, respectively. Two TBQ-2C irradiance meters are fixed on bracket to record the global horizontal irradiance (G_{GHI}). The two irradiance meters are located at the highest and lowest points of BHCPV module respectively.

There are two reasons for choosing an aluminum film with a reflectance of 0.9 in this study. Firstly, the experimental ground is normalized to reduce optical errors caused by ground non-uniformity.

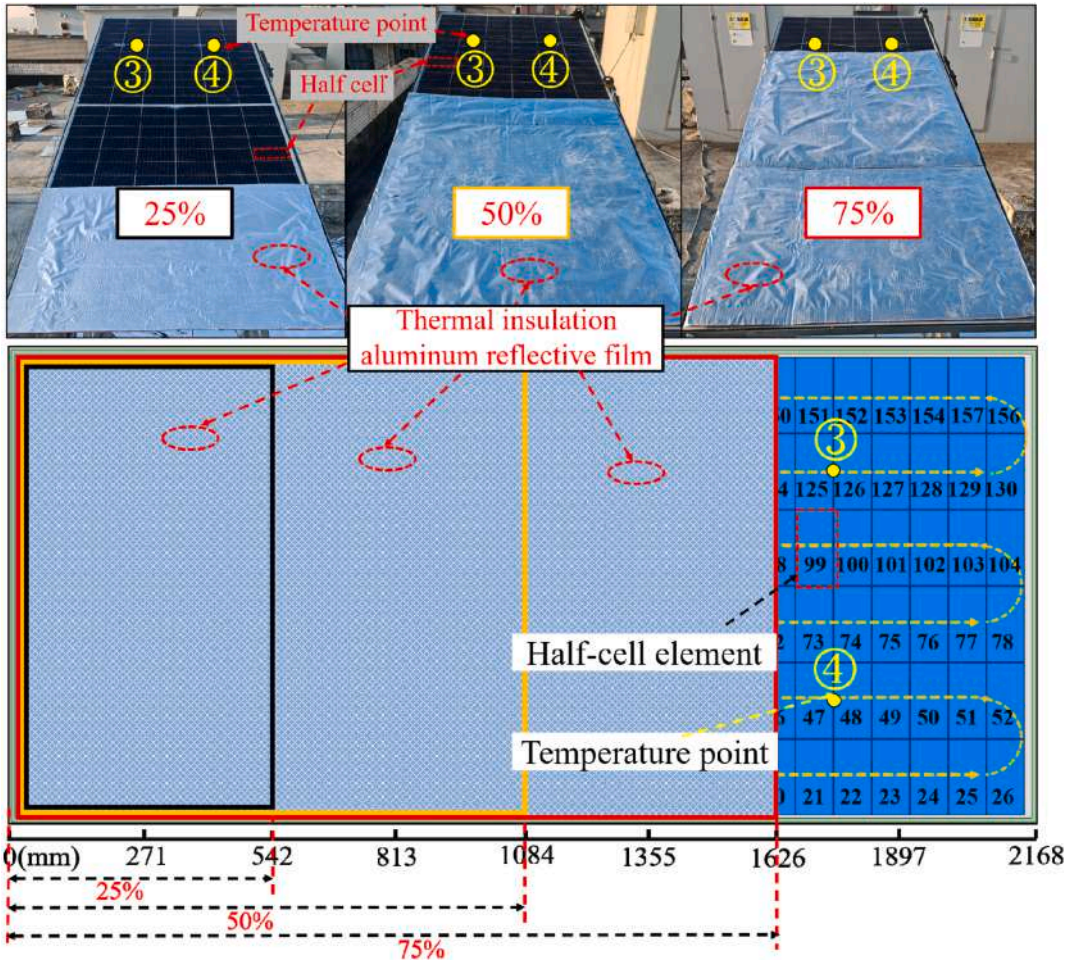


Fig. 7. Horizontal shading under different shading ratios.

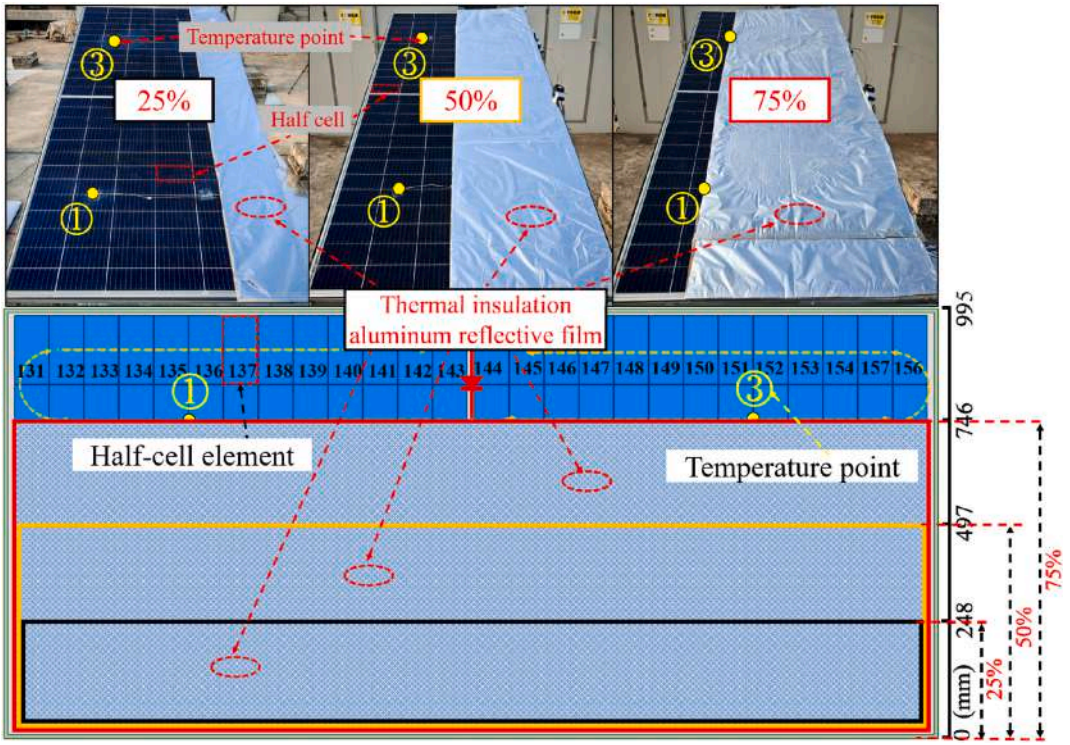


Fig. 8. Vertical shading under different shading ratios.

Secondly, high reflectivity would further exacerbate the non-uniformity of irradiance and temperature in BHCPV module, as well as electrical nonlinearities, especially under shading conditions. This testing environment can more acutely expose the potential ability of IM-DEC model when simulating complex coupling effects, and can better verify the correctness and applicability of IM-DEC model.

As shown in Fig. 9, in order to obtain the temperature characteristics of BHCPV module, Ten K-type thermocouples are arranged at front and rear sides, where 1~5 are arranged at front side and 6~10 are arranged at rear side. Two K-type thermocouples are used to detect ambient temperature. SN-3000-FSJT-GPRS anemometer and SN-3000-FXJT-N01 wind vane are used to record wind velocity and wind direction. PROVA 210 I-V curve tester is used to measure real-time electrical properties of BHCPV modules. An Agilent data acquisition instrument (Agilent 34970A) and its companion module are used to record real-time data. The parameters of main test instruments are listed in Table 2.

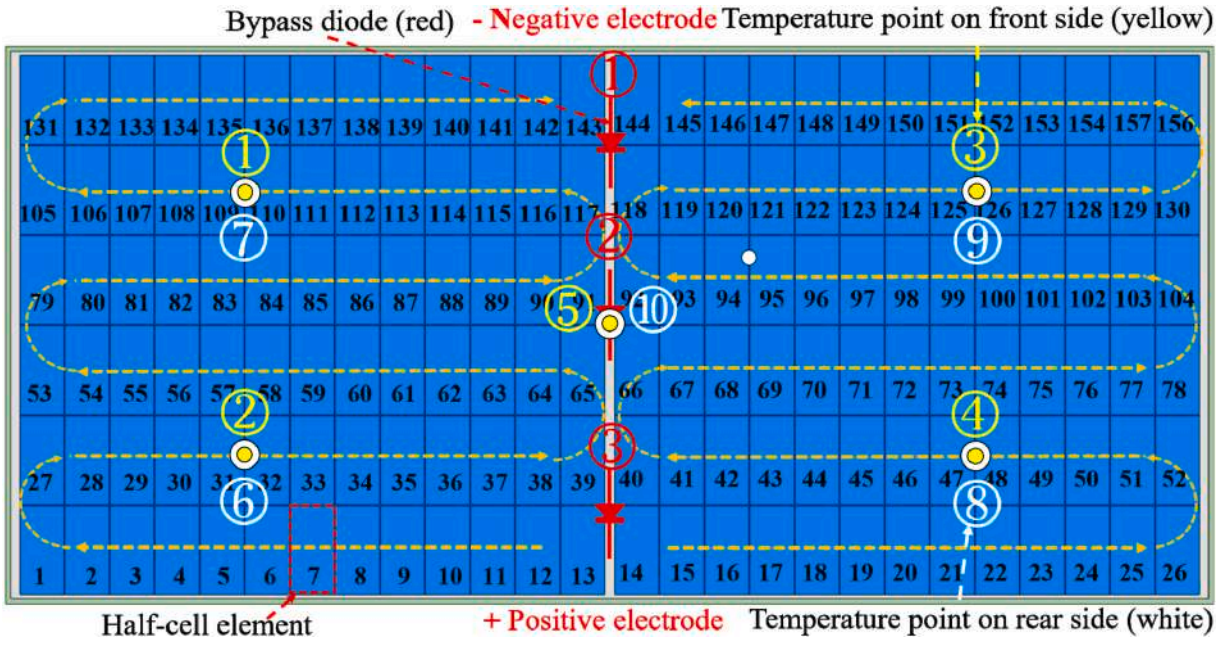


Fig. 9. Temperature measurement points on front and rear sides of BHCPV module.

Table 2
Testing instrument parameters.

Instruments	Ranges	Accuracy
SN-3000-FXJT-N01 Wind Vane	8 directions	-
SN-3000-FSJT-GPRS anemometer	0-70 m/s	±0.1 m/s
PROVA210 I-V curve tester	0-60 V, 0.01-12 A	±1%
Agilent data acquisition instrument (34970A)	0.0002-1000 V, 0-1 A	-
WRNT-TP-SC-06-2PBO K-type thermocouple	0-600 °C	-
TBQ-2C irradiance meter	0-2000 W/m^2	<5%

5.1. Experimental uncertainty analysis and model validation under non-shading

Both instrumental errors and operational errors can significantly influence the experimental results. The relative uncertainty ($\Delta y/y$) associated with key experimental parameters is evaluated based on Holman's uncertainty analysis method [48], as given by the following formula:

$$\frac{\Delta y}{y}=\sqrt{\left(\frac{\partial f}{\partial x_1}\right)^2\left(\frac{\Delta x_1}{y}\right)^2+\left(\frac{\partial f}{\partial x_2}\right)^2\left(\frac{\Delta x_2}{y}\right)^2\cdots+\left(\frac{\partial f}{\partial x_n}\right)^2\left(\frac{\Delta x_n}{y}\right)^2} \tag{26}$$

where f denotes a function of the independent variables x_1, x_2, …., x_n; each x_i represents an input variable of the function f; and Δx_n is absolute error associated with the variables. Current (I), voltage (V) and global horizontal irradiance and temperature are directly measured parameters.

Relative uncertainty in P of BHCPV module:

$$\frac{\Delta P}{P}=\left[\left(\frac{\Delta V}{V}\right)^2+\left(\frac{\Delta I}{I}\right)^2\right]^{\frac{1}{2}}=\left(1\%^2+1\%^2\right)^{\frac{1}{2}}=1.414\% \tag{27}$$

The calculated relative uncertainty of P is below 2%, indicating that the measurement approach employed in this experiment is highly accurate.

To verify the accuracy of IM-DEC model under non-shading, the I-V/P-V curves of BHCPV module are first obtained under STC condition, as shown in Fig. 10(a). The simulated open circuit voltage, short circuit current, maximum power, and voltage and current at the maximum power point are 53.37 V, 10.54 A, 445.14 W, 43.77 V, and 10.17 A, respectively. These values are basically consistent with the data provided by manufacturer (open circuit voltage, short circuit current, maximum power, and voltage and current at the maximum power point

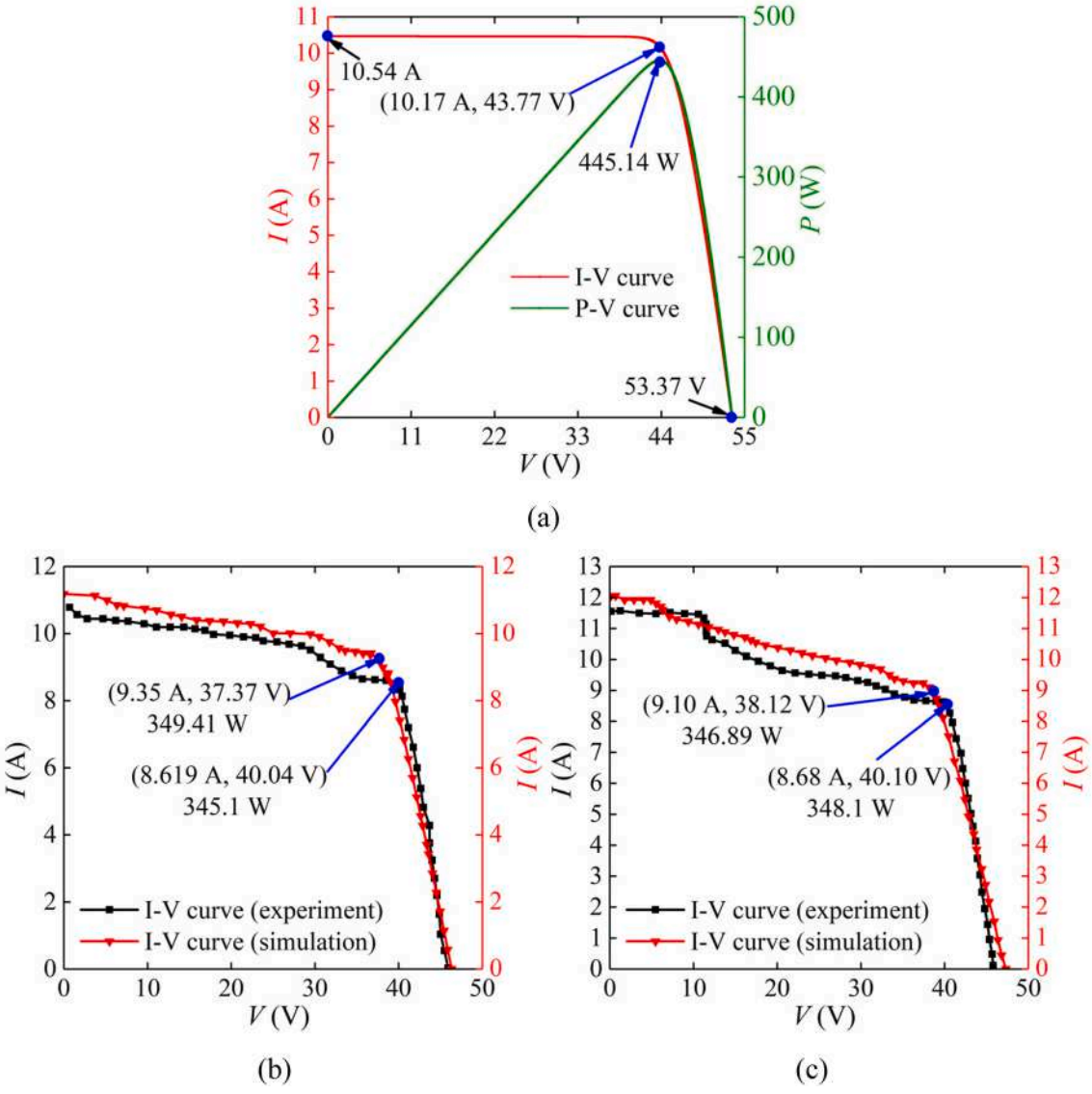


Fig. 10. IM-DEC model verification under non-shading: (a) I-V/P-V curves of BHCPV module under STC conditions; (b) I-V curves obtained from simulation and experiment at 12:00 on July 22, 2024; (c) I-V curves obtained from simulation and experiment at 13:30 on July 22, 2024.

are 53.37 V, 10.52 A, 445 W, 43.93 V, and 10.13 A, respectively), thus verifying the reliability of the circuit model.

It is worth pointing out that, as shown in Fig. 10(a), the I-V/P-V curve is relatively smooth. This is because the curve is not measured under a real outdoor varying environment, but is the ideal steady-state simulation results calculated based on IM-DEC model under standard test conditions (STC: 1000 W/m^2, 25 °C). Under STC, $G_{e,i}$ and $t_{ave,i}$ input to each HCEs in the simulation are set to the same and constant values, ignoring environmental perturbations such as irradiance fluctuations and temperature transients. Therefore, the simulation curve shows a continuous and smooth shape, which is the normal output performance of the model under ideal input, and it matches well with the key electrical parameters provided by the manufacturer under STC. In addition, the I-V curve of BFCPV module shown in Refs. [25,49] under STC also exhibits a smooth shape, further confirming the rationality of the smooth curve in Fig. 10(a). It should be emphasized that the main purpose of verification under STC is to serve as a benchmark for circuit topology and parameter verification, rather than to reflect the fluctuation characteristics caused by environmental perturbations in outdoor measured curves.

To further verify the accuracy of IM-DEC model under non-standard conditions (i.e., environmental fluctuations), the I-V curves of BHCPV module measured at 12:00 and 13:30 on July 22, 2024, are compared and verified. The simulation and experimental results are shown in Fig. 10(b) and (c), respectively. Detailed information about experiment can be found in Ref. [46]. These figures show that the I-V curves obtained from both experiment and simulation exhibit some fluctuations. This is mainly due to two reasons. Firstly, the partially cloudy weather during the test period causes instantaneous irradiance fluctuations, affecting the stability of experimental I-V curves. Secondly, the non-uniformity of irradiance and temperature distribution on module surfaces also causes fluctuations in curves, and similar conclusions are reached in Refs. [39,47]. Specifically, at 12:00 on July 22, the relative errors between the P obtained from simulation and experiment, and the corresponding current and voltage, as well as the root mean square error (RMSE) of current in I-V curve, are 1.25%, 6.67%, 8.48%, and 0.53 A, respectively; while at 13:30, the relative errors of the above parameters are 0.33%, 4.94%, 4.84%, and 0.56 A, respectively. For error analysis, Section 5.5 will be given.

5.2. IM-DEC model validation under horizontal shading

The experimental results of seven moments from 10:00 to 16:00 on September 13, 2024 are selected to verify the accuracy of IM-DEC model under different shading ratios during horizontal shading and compare with CO-DEC and AVT models. In the following studies, the subscripts "exp" and "sim" represent experimental and simulated values, respectively. Fig. 11(a) shows ambient temperature (t_{amb}), wind velocity and wind direction at different moments. Fig. 11(b) shows global horizontal irradiance ($G_{GHI,exp}$) obtained by two TBQ irradiance meters at different moments, and global horizontal irradiance ($G_{GHI,sim}$) and direct normal irradiation ($G_{DNI,sim}$) calculated based on Refs. [41,42]. The maximum relative errors of $G_{GHI,sim}$ and $G_{GHI,exp}$ are less than 10.84% and the average relative error is 7.41%, which proves the correctness of the calculation method from Refs. [41,42].

$G_{DNI,sim}$ will be used as input parameters for direct solar radiation model and combined with sky diffuse radiation model to perform optical simulations and obtain the irradiance distribution on front and rear sides of BHCPV module. The optical results are used as irradiance input parameters for IM-DEC, CO-DEC and AVT models.

Fig. 12 shows the difference between simulated maximum instantaneous electric power (P_{sim}) and experimental values (P_{exp}) of BHCPV modules under various horizontal shading ratios, as calculated using AVT, CO-DEC, and IM-DEC models. From Fig. 12(a) and (b), the relative errors between P_{sim} calculated using AVT model and P_{exp} are relatively small at shading ratios (ζ_f) of 0% and 50%, with average relative error of 6.04% and 6.06%, respectively. In contrast, significantly larger errors are observed at $\zeta_f = 25\%$ and 75%, with average relative error of 29.21% and 107.15%.

To better explain the causes of above phenomenon, an analysis will be conducted in combination with Figs. 2, 7 and 12(a) and (b). When ζ_f = 0%, there is non-shading on BHCPV module, the irradiance and temperature distribution are approximately uniform, and all strings work normally. Under this condition, P is estimated based on AVT model by using average irradiance and average temperature as input parameters. This approach can reflect the operating conditions of module under non-shading conditions, resulting in a relatively small deviation between P_{sim} and P_{exp}.

Since BHCPV module adopts a symmetrical configuration, where the upper and lower strings are connected in parallel and sharing a common bypass diode, its electrical response exhibits significant variations under different ζ_f. When ζ_f = 25%, shading begins from the lower part of

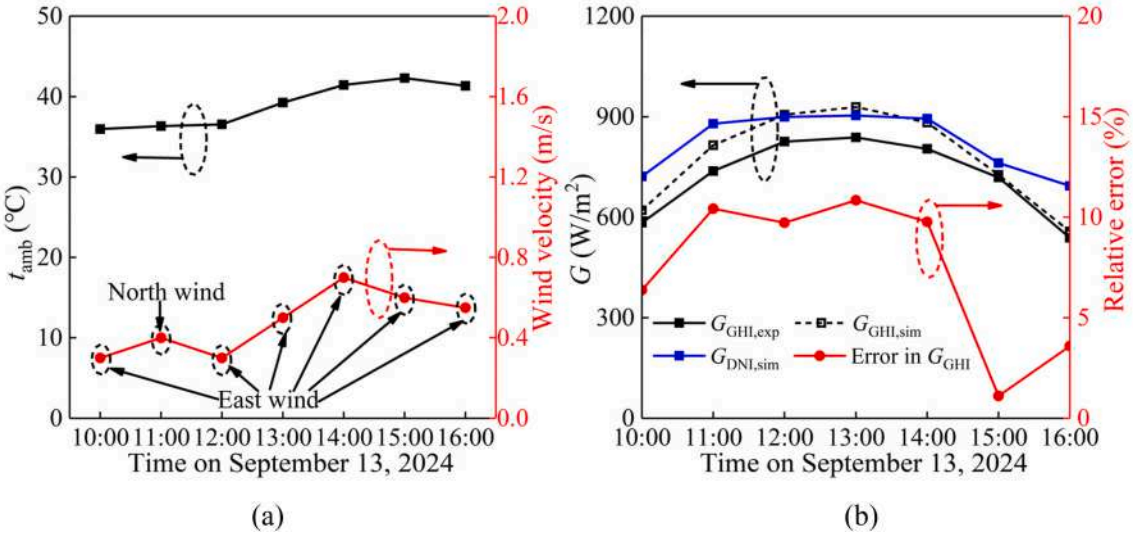


Fig. 11. Environmental parameters: (a) Ambient temperature (t_{amb}), wind velocity and wind direction; (b) Global horizontal irradiance obtained by experiment ($G_{GHI,exp}$) and calculation ($G_{GHI,sim}$), direct normal irradiation ($G_{DNI,sim}$) obtained by calculation.

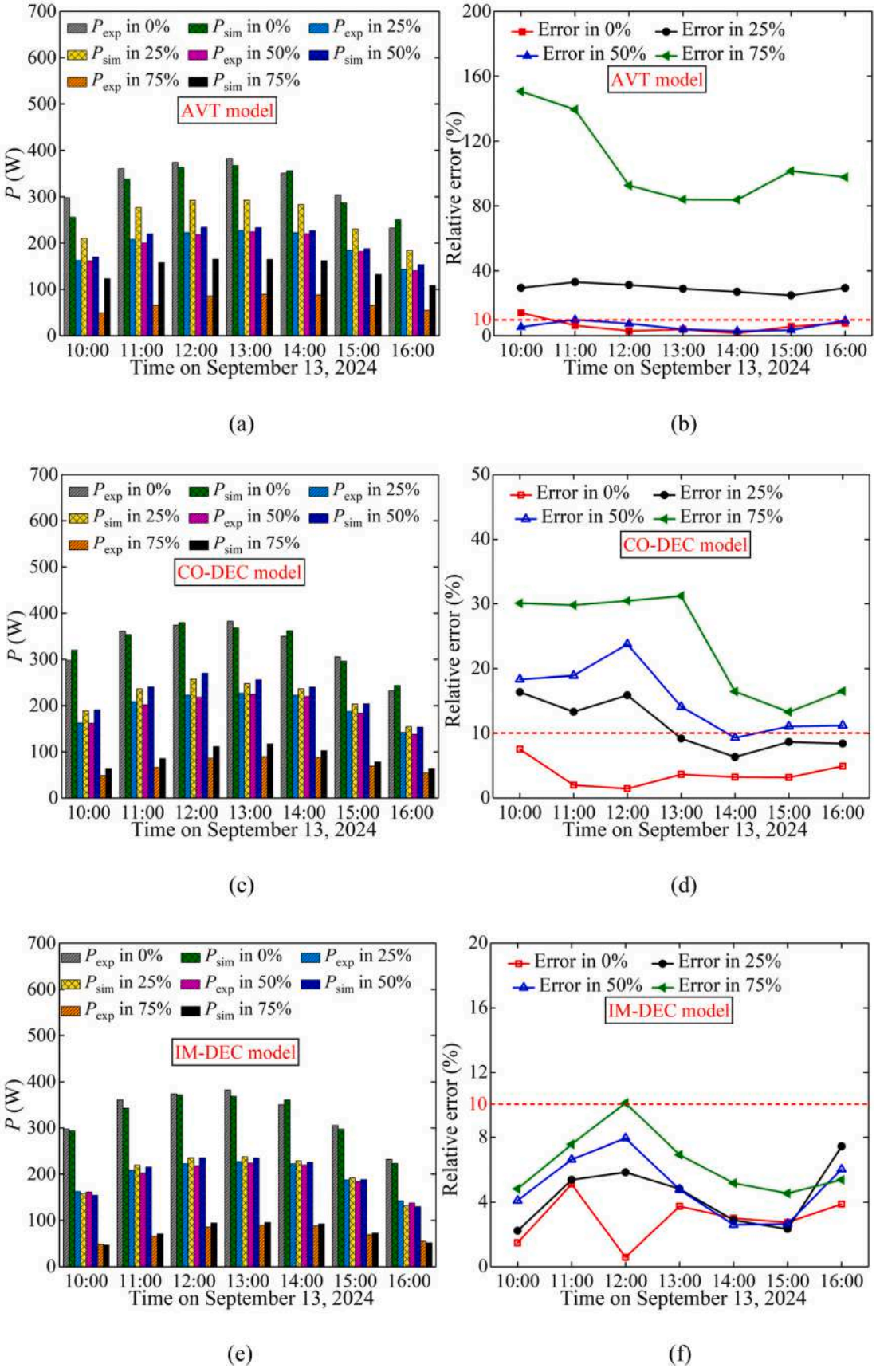


Fig. 12. Comparison of the maximum instantaneous electric power (*P*) under simulations and experiment at different horizontal shading ratios: (a) AVT model; (b) Relative error for AVT model; (c) CO-DEC model; (d) Relative error for CO-DEC model; (e) IM-DEC model; (f) Relative error for IM-DEC mode.

BHCPV module, resulting in half of the lower area being shaded. Most of HCEs in strings 4 to 6 enter reverse bias. At this point, the main current is supplied by strings 1 to 3, and the module shows a nonlinear electrical response. However, AVT model is based on the principle of energy conversion, assuming a uniform reduction in front side irradiance to 75% of non-shading condition. It assumes that all strings are functioning normally, thereby failing to account for electrical non-uniformity within the module. In particular, it neglects the nonlinear response caused by localized mismatch. As a result, P_{sim} is significantly overestimated compared to P_{exp}.

Similarly, when $\zeta_f = 50\%$, AVT model assumes that front side irradiance is reduced to 50% of non-shading condition, and calculates the module output power accordingly. Simulation results indicate that when front side irradiance is reduced by 50%, P_{sim} drops to approximately 60% of non-shading condition, due to the bifacial characteristics of BHCPV module. It should be pointed out that P_{sim} obtained by AVT model closely matches P_{exp} under the condition of $\zeta_f = 50\%$. However, this agreement is specific to the electrical configuration of tested BHCPV module, that is, the module can achieve normal operation of the non-shading strings when the upper or lower strings are completely shaded. This ultimately leads to an illusion that the results calculated using AVT model for $\zeta_f = 50\%$ are close to experimental results. If the module uses a different internal interconnection scheme or shading patterns, the model may not accurately capture the actual performance. For example, when the vertical shading ratio is 50%, the relative error is large, as detailed in Section 5.3.

In the case of $\zeta_f = 75\%$, strings 4 to 6 contribute nothing to power output, while half of the front side area of strings 1 to 3 is shaded. As a result, the system operates under a highly non-uniform electrical condition. Consequently, the relative error between P_{sim} calculated by AVT model and P_{exp} becomes significantly larger.

Fig. 12(c) and (d) show the difference between P_{sim} calculated by CO-DEC model and P_{exp} under various horizontal shading ratios. As shown in Fig. 12(d), as ζ_f increases from 0% to 75%, the average relative errors gradually increase to 3.72%, 11.18%, 15.26% and 23.99%, respectively. This is because, with increasing ζ_f, the nonlinear characteristics of the internal electrical response in module become more significant. When ζ_f is larger, bypass diodes are activated, leading to a reconfiguration of the current path. These effects gradually diminish the accuracy of CO-DEC model, which is based on the assumption of uniform average input conditions. Besides, the simplified treatment of irradiance and the assumption of uniform temperature contribute to the accumulation of error, causing the average relative error increase with the increase of ζ_f. This trend reflects the limitations of CO-DEC model under complex shading conditions.

To address the shortcomings of CO-DEC model, IM-DEC model is employed to calculate electrical performance, the results are shown in Fig. 12(e) and (f). This model account for non-uniformity of irradiance and temperature on HCEs and introduces a dynamic thermal-electric coupling mechanism. Fig. 12(f) shows that the average relative errors between P_{sim} calculated by IM-DEC model and P_{exp} are 2.93%, 4.41%, 5.00%, and 6.35%, respectively, under different ζ_f conditions. The overall computational accuracy is better than that of AVT and CO-DEC models.

In addition, to further reveal the internal circuit response mechanism of BHCPV module under horizontal shading conditions, Fig. 13(a) and (b) show the I-V/P-V curves under various horizontal shading ratios at 13:00 on September 13, 2024 based on IM-DEC model simulation.

As can be seen from Fig. 13(a) and (b), when $\zeta_f = 0$, each string operates normally, and three bypass diodes are in the off state. I-V/P-V curves are only slightly disturbed by NUIR and NUT in medium-high current region, and generally exhibit typical single peak characteristics.

When $\zeta_f = 25\%$ and 50%, strings 4 to 6 are partially and completely shaded, while strings 1 to 3 receive light normally. Since BHCPV module adopts the same hybrid topology as in Ref. [34], near the short circuit point and maximum power point, most of shaded HCEs enter the reverse bias state. The equivalent current at module end is almost entirely provided by the non-shading strings 1 to 3, and the contribution of strings 4 to 6 to external current is extremely limited, only bearing a limited reverse current. The bypass diodes are still in the off state. From a macroscopic perspective, the module is equivalent to a structure of "only three non-shading strings connected in series". Therefore, when ζ_f is 25% and 50%, the short circuit current and P of module are stable at about 60% of non-shading condition, and I-V/P-V curves always maintain a single peak shape. This phenomenon is consistent with the conclusion in Ref. [34].

When $\zeta_f = 75\%$, there are no longer any complete non-shading strings inside module. A large number of shaded HCEs operate in reverse bias, and the module can be equivalent to a topology circuit of "a few non-shading HCEs + multiple bypass branches". The short circuit current and maximum output power continue to decrease. However, due to the similar light-receiving conditions and current-limiting levels of each parallel group at this stage, the voltage thresholds for entering bypass conduction or deep reverse conduction are also relatively close.

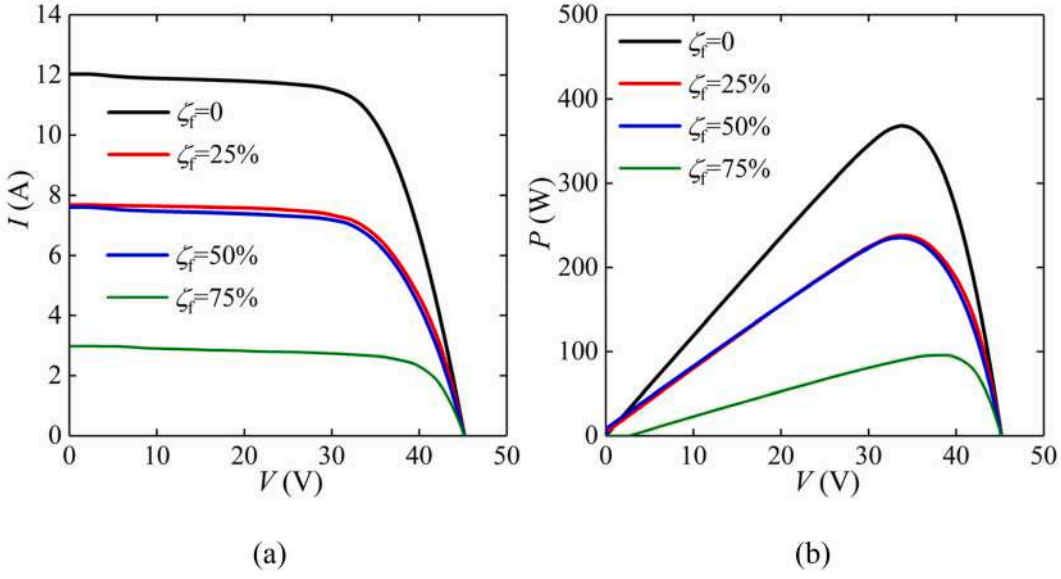


Fig. 13. I-V/P-V curves under different horizontal shading ratios at 13:00 on September 13, 2024 based on IM-DEC model simulation: (a) I-V curve; (b) P-V curve.

Multiple local circuit reconfiguration processes are superimposed and "smoothed out" in the macroscopic I-V curve. Consequently, I-V/P-V curves still only show a smooth main knees and a single power peak, which is consistent with the result in Ref. [34].

In summary, the fundamental reason why the I-V/P-V curves of BHCPV module under horizontal shading conditions always exhibit a single-peak shape is not that "the step is not visible due to insufficient shading", but rather that when ζ_f reaches 25% or higher, most of HCEs in shading strings 4 to 6 have entered reverse bias within the main operating area, severely limiting the current within the strings. The positive contribution of strings 4 to 6 to the parallel group port current is extremely limited, while the module loop current is mainly provided by strings 1 to 3. Therefore, at $\zeta_f = 25\%$ and 50%, P is similar, exhibiting a current-limited behavior determined by strings 1 to 3. That is, once the shading strings are electrically marginalized, further increasing the shading ratio will not proportionally reduce the module output. After exceeding the shading ratio $\zeta_f = 50\%$, the further downward shift of curve at $\zeta_f = 75\%$ corresponds to the continued reduction in the number of non-shading HCEs.

The differences of simulated temperature measurement point 3 ($t_{3,sim}$) and point 4 ($t_{4,sim}$) from the experimental values are obtained through optical-thermal-electrical-environmental multiphysics calculations based on IM-DEC model. The results are illustrated in Fig. 14(a) and (b). Calculations show that the maximum and average relative errors between $t_{3,sim}$ and $t_{3,exp}$ are less than 6.22% and 2.63%, respectively, and those between $t_{4,sim}$ and $t_{4,exp}$ are less than 10.23% and 5.26%, respectively. Similar validation results are observed at other measurement points.

Additionally, taking 13:00 as an example, the temperature distribution on BHCPV module and the characteristics of flow field around module under different horizontal shading ratios are shown in Fig. 15(a) and (b), respectively. It should be noted that since the flow fields under different ζ_f are similar, only the flow field at $\zeta_f = 25\%$ is provided, as shown in Fig. 15(b). Similar results are also found at other times. From Fig. 15(a), due to the combined effects of ambient wind, buoyancy forces and partial shading, a NUT distribution appears on module surface, with lower temperatures in shaded regions and higher temperatures in non-shading regions. Notably, even adjacent HCEs within non-shading region can exhibit substantial temperature differences. These findings highlight the necessity of employing IM-DEC model with dynamic thermal-electrical coupling for each HCE.

5.3. IM-DEC model validation under vertical shading

To further evaluate the accuracy of IM-DEC model under different vertical shading ratio, the experimental results at seven moments from 10:00 to 16:00 on September 14, 2024 are selected for validation and compared with CO-DEC and AVT models. The environmental parameters at different moments are shown in Fig. 16(a). $G_{GHI,sim}$ and $G_{DNI,sim}$ are calculated according to Refs. [41,42] and the results are shown Fig. 16(b). The maximum relative errors of $G_{GHI,sim}$ and $G_{GHI,exp}$ are less than 11.64%, and the average relative error is 5.18%, which proves the correctness of the calculation method of Refs. [41,42] again.

Fig. 17(a) and (b) demonstrate the difference between P_{sim} calculated by AVT model and P_{exp} for different ζ_f under vertical shading. Fig. 17(b) shows that the average relative errors are 4.50%, 26.18%, 91.49%, and 90.81% when ζ_f is 0%, 25%, 50%, and 75%, respectively. The reason for the errors when $\zeta_f = 0\%$ can be found above. According to Figs. 2, 8 and 17(a) and (b), it can be seen that at $\zeta_f = 25\%$, shading begins from the right side of BHCPV module, HCEs 40~52 in string 3 and HCEs 27~39 in string 6 are partially shaded. As a result, bypass diode 3 is activated, and the electrical output is primarily contributed by strings 1, 2, 4, and 5. Similarly, when $\zeta_f = 50\%$, bypass diodes 2 and 3 are activated, and the electrical output is primarily supported by strings 1 and 4. At $\zeta_f = 75\%$, the front sides of strings 2, 3, 5, and 6 are completely shaded, while HCEs 118~130 in string 1 and HCEs 105~117 in string 4 are partially shaded. Therefore, the electrical output decreases sharply. In this case, AVT model still assumes that all strings operate normally and only considers the change in front side irradiance, ignoring the circuit reconstruction triggered by local mismatches, which results in P_{sim} being significantly higher than P_{exp}.

In addition, combining Figs. 12(b) and 17(b), for $\zeta_f = 50\%$, a significant difference is observed between the relative errors under horizontal and vertical shading conditions as calculated by AVT model. This discrepancy is mainly attributed to the different electrical responses induced by the two shading patterns. When ζ_f in horizontal shading is 50%, the electrical output is mainly borne by strings 1 to 3, which can be interpreted as the effective power generation area on front side of module being halved, resulting in a small difference between P_{sim} and P_{exp}. However, under vertical shading conditions with $\zeta_f = 50\%$, only strings 1 and 4 are fully illuminated, while the other strings are either fully or partially shaded. The electrical response is different from that under horizontal shading conditions, resulting in a larger relative error.

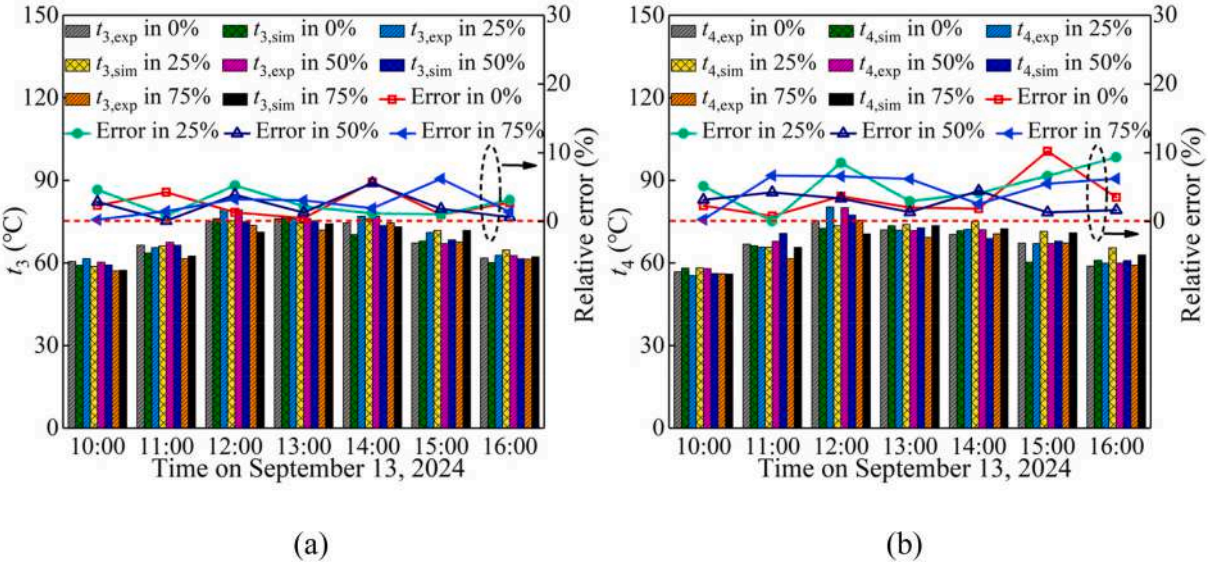


Fig. 14. Comparison of temperature between simulation and experiment at different horizontal shading ratios: (a) Temperature measured at point 3; (b) Temperature measured at point 4.

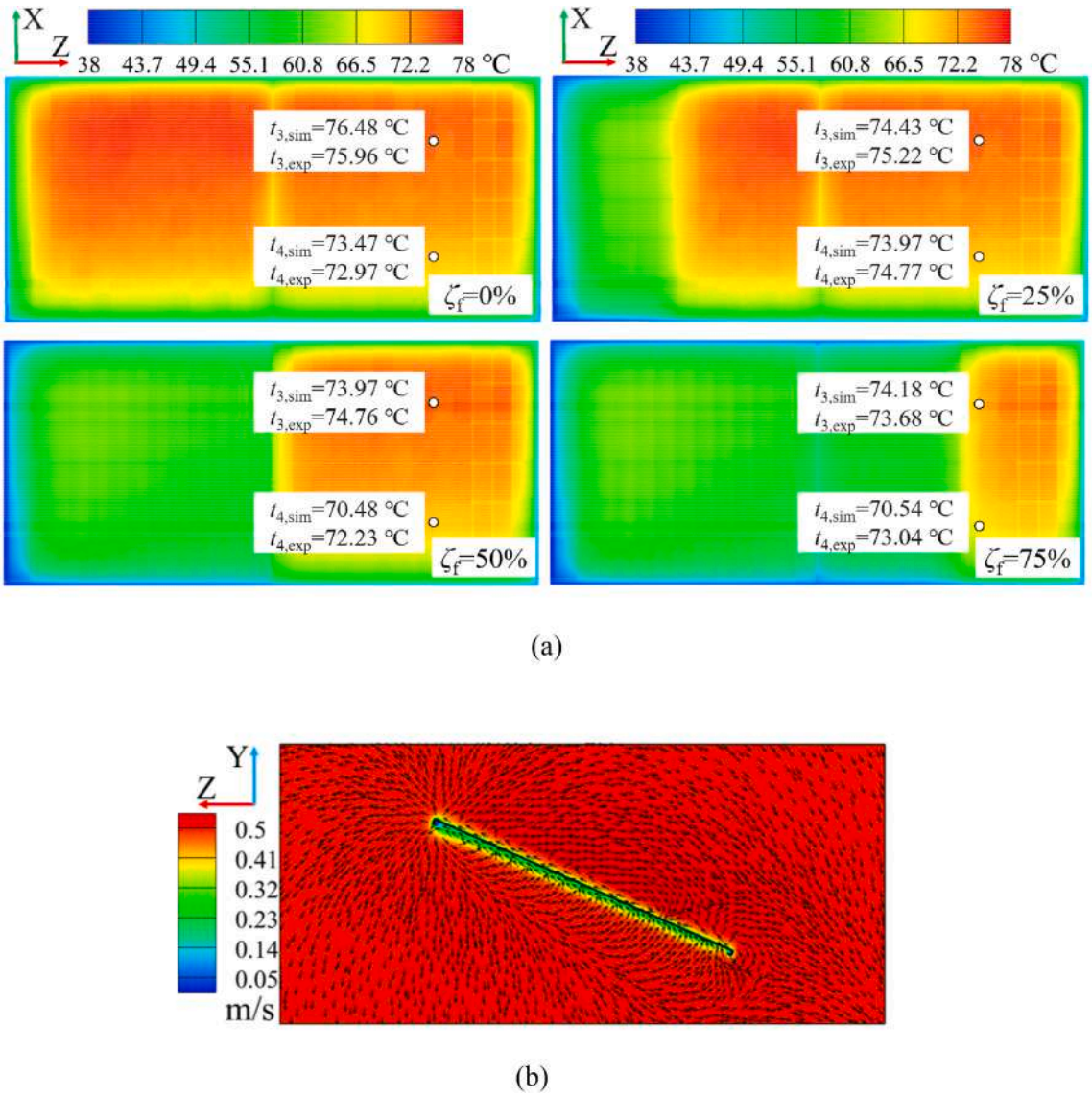


Fig. 15. Temperature and flow fields characteristics at 13:00 under different horizontal shading ratios: (a) Temperature distribution on BHCPV modules; (b) Flow field around BHCPV module.

The differences between P_{sim} calculated using CO-DEC model and P_{exp} under different ζ_f are illustrated in Fig. 17(c) and (d). As ζ_f increases, the average relative error are 8.29%, 11.17%, 12.33% and 25.72%, respectively. The analysis of the errors can be found above. The differences between P_{sim} calculated using IM-DEC model and P_{exp} under different ζ_f are shown in Fig. 17(e) and (f). Fig. 17(f) shows that the average relative errors under different ζ_f are 4.69%, 5.86%, 6.43%, and 6.42%, respectively. The relative error exceeds 10% under only two specific operating condition, and the overall computational accuracy is better than that of AVT and CO-DEC models.

In addition, to further reveal the internal circuit response mechanism of BHCPV module under vertical shading conditions, Fig. 18(a) and (b) show the I-V/P-V curves under various vertical shading ratios at 13:00 on September 14, 2024 based on IM-DEC model simulation.

When $\zeta_f = 0$, the phenomenon is consistent with that in Fig. 13 (the horizontal shading ratio is 0), and I-V/P-V curves generally exhibits typical single peak characteristics.

When ζ_f is 25% and 50%, the shading starts from the long side on right and extends to left along the short side. At this time, strings 3 and 6, as well as strings 2 and 5 which are subsequently included in shading area, have a large number of fully and partially shaded HCEs. Strings 1 and 4 are non-shading when ζ_f = 25% and remain in a relatively "dominant string" state when $\zeta_f = 50\%$. During the downward scan from open circuit voltage, the shaded HCEs in strings 3, 6, 2 and 5, are the first to enter the reverse bias state. This causes strings 3, 6, 2 and 5 to gradually change from "participating in power generation" to "being taken over by bypass branch". Macroscopically, I-V curve shows a distinct current step in the medium-to-high voltage region, while P-V curve exhibits a multi-peak structure. As ζ_f increases from 25% to 50%, the number of shaded HCEs increases, causing the main peak of P-V curve to shift significantly downward and the secondary peaks to rise relatively. P decreases from 415.69 W at $\zeta_f = 0$ to 116.71 W at $\zeta_f = 50\%$.

When $\zeta_f = 75\%$, strings 2, 3, 5, and 6 are completely shaded, while only half of the HCEs in strings 1 and 4 are normally illuminated. There are no longer any complete non-shading "dominant strings" within module. A large number of shaded HCEs remain in a reverse bias state within the main operating range, and the module can be equivalent to a structure of "a few non-shading HCEs + multiple bypass branches". I-V curve initially shows a current step dominated by non-shading HCEs in high-voltage region, corresponding to a low and wide main peak in P-V curve. As the voltage continues to decrease, all HCEs almost simultaneously enter a mismatch state. However, knee point and secondary

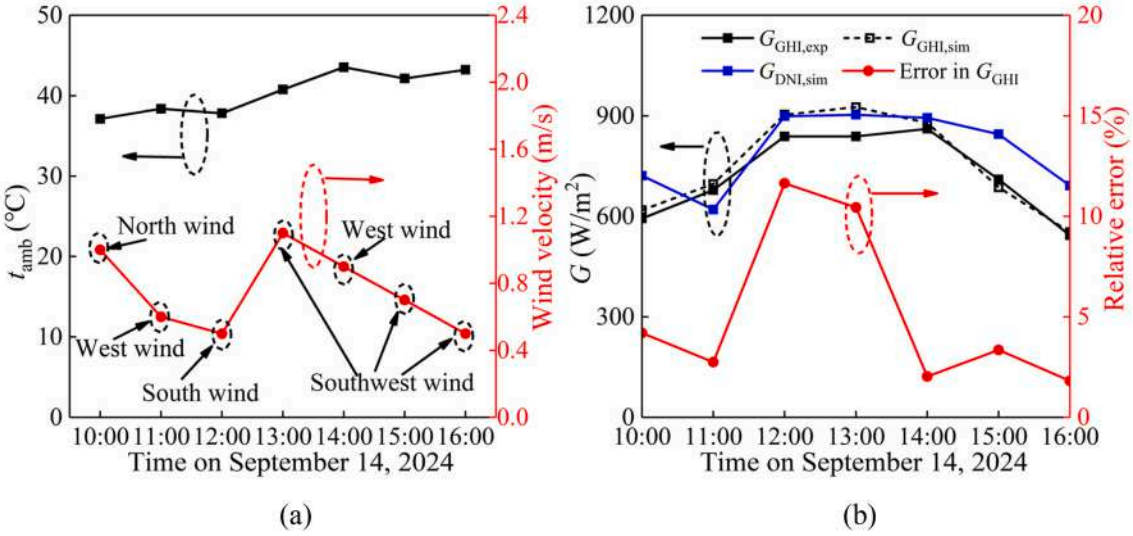


Fig. 16. Environmental parameters: (a) Ambient temperature (t_{amb}), wind velocity and wind direction; (b) Global horizontal irradiance obtained by experiment ($G_{GHI,exp}$) and calculation ($G_{GHI,sim}$), direct normal irradiation ($G_{DNI,sim}$) obtained by calculation.

peak can still be observed in curve, although their power is much lower than the high-voltage main peak, with P only reaching 103.3 W.

In summary, as ζ_f increases from 0 to 75%, I-V/P-V curves of the module gradually evolves from a single peak under non-shading conditions to a "multi-step, multi-peak" form. The evolution trend is similar to the electrical characteristics of module A under vertical shading conditions in Ref. [34].

Fig. 19(a) and (b) present the difference between the simulated temperature measurement points 1 ($t_{1,sim}$) and 3 ($t_{3,sim}$) and experimental values. The results show that the maximum and average relative errors between $t_{1,sim}$ and $t_{1,exp}$ are less than 10.58% and 6.77%, respectively, and those between $t_{3,sim}$ and $t_{3,exp}$ are less than 10.33% and 7.87%, respectively. Similar validation results are observed at other measurement points.

Additionally, taking 13:00 as an example, the temperature distribution on BHCPV module and the characteristics of flow field around module under different vertical shading ratios are shown in Fig. 20(a) and (b), respectively. Only the flow field at $\zeta_f = 25\%$ is provided, as shown in Fig. 20(b). Similar results are also found at other times. From Fig. 20(a), BHCPV module exhibits obvious NUT distribution. Consequently, assuming uniform irradiance and temperature distributions in AVT and CO-DEC models results in calculation errors, especially when the shading ratio is large.

Temperature field simulation results show that significant temperature differences may exist between adjacent HCEs in both shaded and non-shading areas. This phenomenon stems from two primary reasons: firstly, NUIR distribution leads to NUT distribution; secondly, the complex flow field exacerbates the non-uniformity of temperature distribution. Furthermore, it is worth noting that under complex shading conditions (such as diagonal, random, or time-varying shadings), reverse bias may lead to increased heat generation. To accurately simulate the thermal process under these conditions, incorporating the reverse bias heating effect into heat source term of IM-DEC model can enhance the model's ability to predict potential hotspot formation or early degradation regions. However, this improved model still needs further validation under steady-state experimental conditions, which will be the focus of future work.

In conclusion, under both horizontal and vertical shading, the prediction accuracy of IM-DEC model for the electrical performance of BHCPV module under different shading ratios is higher than that of AVT and CO-DEC models. Its advantages are mainly reflected in the following two aspects.

Regarding the irradiance input mechanism: IM-DEC model accurately captures the impact of local mismatch on electrical performance by establishing NUIR distribution under partial shading at HCE-level. Although CO-DEC model also uses HCE as modeling unit, it applies an averaged irradiance input and ignores the effect of NUIR between HCEs on the electrical response. AVT model assumes that the circuit operates normally under shading conditions, only the intensity of irradiation changes. However, its energy-based electrical performance evaluation cannot capture the actual electrical characteristics of the module.

Regarding the temperature input mechanism: IM-DEC model has a similar solution structure to CO-DEC model. However, the difference is that IM-DEC model obtains the independent average temperature of each HCE based on thermal model and uses it as an input parameter for electrical model. This approach allows IM-DEC model to fully capture the impact of temperature gradient differences within the element on its electrical response. In contrast, CO-DEC model assigns a uniform temperature to all HCEs, ignoring the effect of temperature non-uniformity on electrical performance. AVT model is based entirely on the overall average temperature input, which results in lower accuracy.

5.4. I-V characteristics verification of IM-DEC model based on concrete ground

Considering the actual situation, a flat concrete ground (with a reflectivity of about 0.4) is used as the experimental site to conduct performance experiments on BHCPV module under different shading patterns and shading ratios, especially the I-V curve, to further verify the accuracy of IM-DEC model.

The experiment is conducted on November 20, 2025, from 13:00 to 14:00 (sunny, stable environmental conditions) to explore the I-V characteristics of BHCPV modules with horizontal and vertical shading ratios of 25%, 50%, and 75%. The experimental platform is the same as described above, except that the aluminum film is replaced with a flat concrete ground. The simulation and experimental conditions are kept consistent. The I-V curves of each module under different shading conditions are obtained using IM-DEC model and compared with experimental results, as shown in Fig. 21(a)–(f). The RMSE of the current in I-V curves under different horizontal shading ratios are 0.24 A, 0.34 A, and 0.11 A, respectively, and the relative errors of the P are 1.90%, 1.72%, and 2.16%, respectively. Similarly, the RMSE of the current under different vertical shading ratios are 0.49 A, 0.30 A, and 0.46 A, respectively, with relative errors of 1.52%, 1.01%, and 2.11% for P.

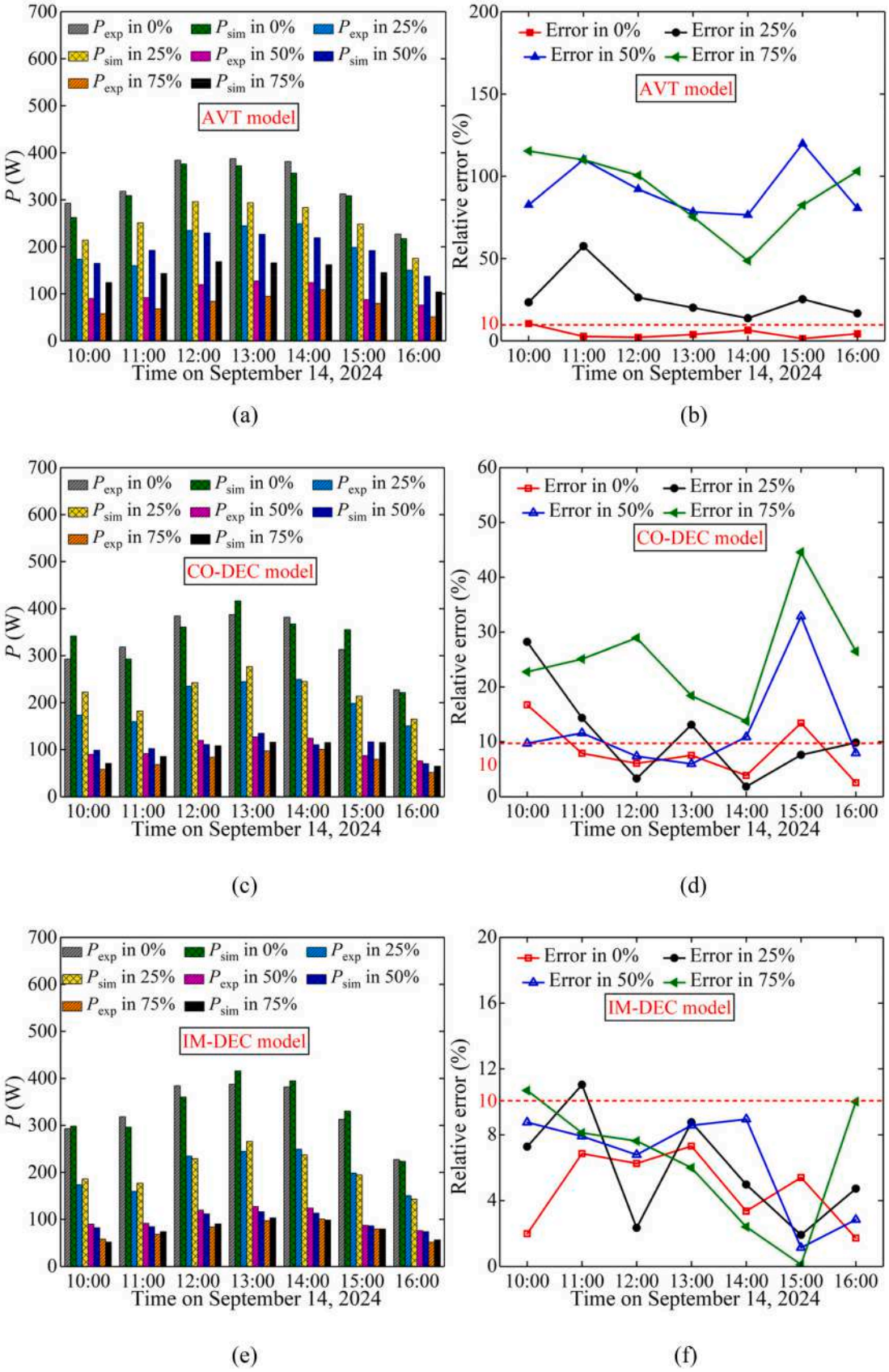


Fig. 17. Comparison of the maximum instantaneous electric power (*P*) between simulation and experiment at different vertical shading ratios: (a) AVT model; (b) Relative error for AVT model; (c) CO-DEC model; (d) Relative error for CO-DEC model; (e) IM-DEC model; (f) Relative error for IM-DEC mode.

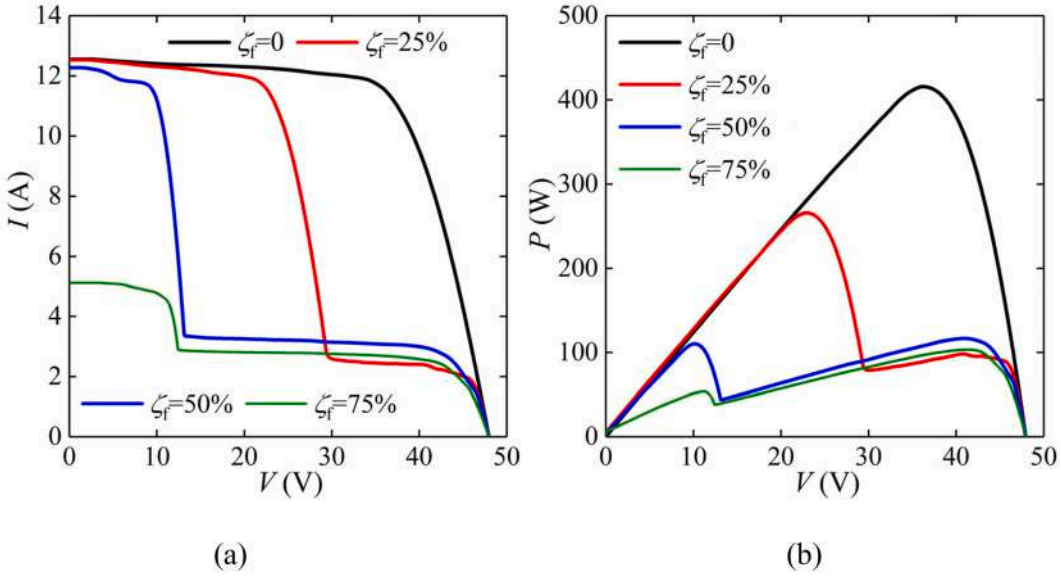


Fig. 18. I-V/P-V curves under different vertical shading ratios at 13:00 on September 14, 2024 based on IM-DEC model simulation: (a) I-V curve; (b) P-V curve.

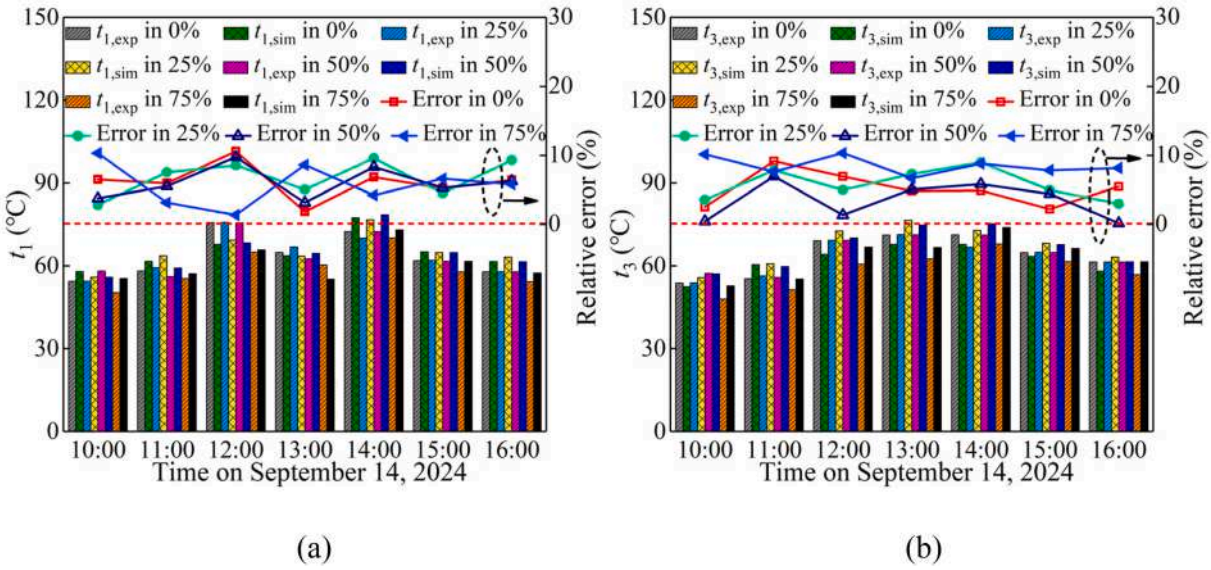


Fig. 19. Comparison of temperature between simulation and experiment at different vertical shading ratios: (a) Temperature measured at point 1; (b) Temperature measured at point 3.

These results verify the correctness of IM-DEC model under different shading conditions. Furthermore, the results show that as the reflectivity decreases, the total irradiance received on rear side decreases, uniformity improves, and the error with the experimental results is lower than that under high reflectivity conditions.

It needs to be emphasized once again that this study primarily focuses on horizontal and vertical shadings because these shadings are standardized baseline cases that can systematically trigger bypass diodes and generate distinct non-uniform patterns, effectively validating the correctness of IM-DEC model. Furthermore, Refs. [16,27,34] also employ standardized horizontal and vertical shadings when studying the impact of shading on PV performance. IM-DEC model has the ability to handle more complex shading because it uses 156 independent HCEs as basic elements for dynamic thermal-electrical coupling calculations. For diagonal and randomly distributed patchy shading, this model can accurately calculate the equivalent irradiance based on the proportion of shaded area for each HCE. For dynamic shading or time-varying shading caused by cloud movement or irradiance fluctuations, this model can capture its dynamic optical-thermal-electrical response through unsteady-state coupling iteration. Therefore, although this study focuses on standard shading patterns, the underlying architecture of IM-DEC model has laid a solid foundation for simulating complex, irregular, and time-varying shadings in real-world applications.

5.5. Error analysis

The aforementioned errors mainly originate from two aspects: firstly, the systematic errors inherent in test instruments themselves; and secondly, the cumulative errors resulting from the coupling effects of multiple physics fields (light, heat, electricity, and environment) during the simulation process. The main error sources will be analyzed in detail below from three levels: optical, thermal, and electrical.

Optical errors: This is the primary source of systematic errors. Firstly, the irradiance meter exhibits response hysteresis under fluctuating solar

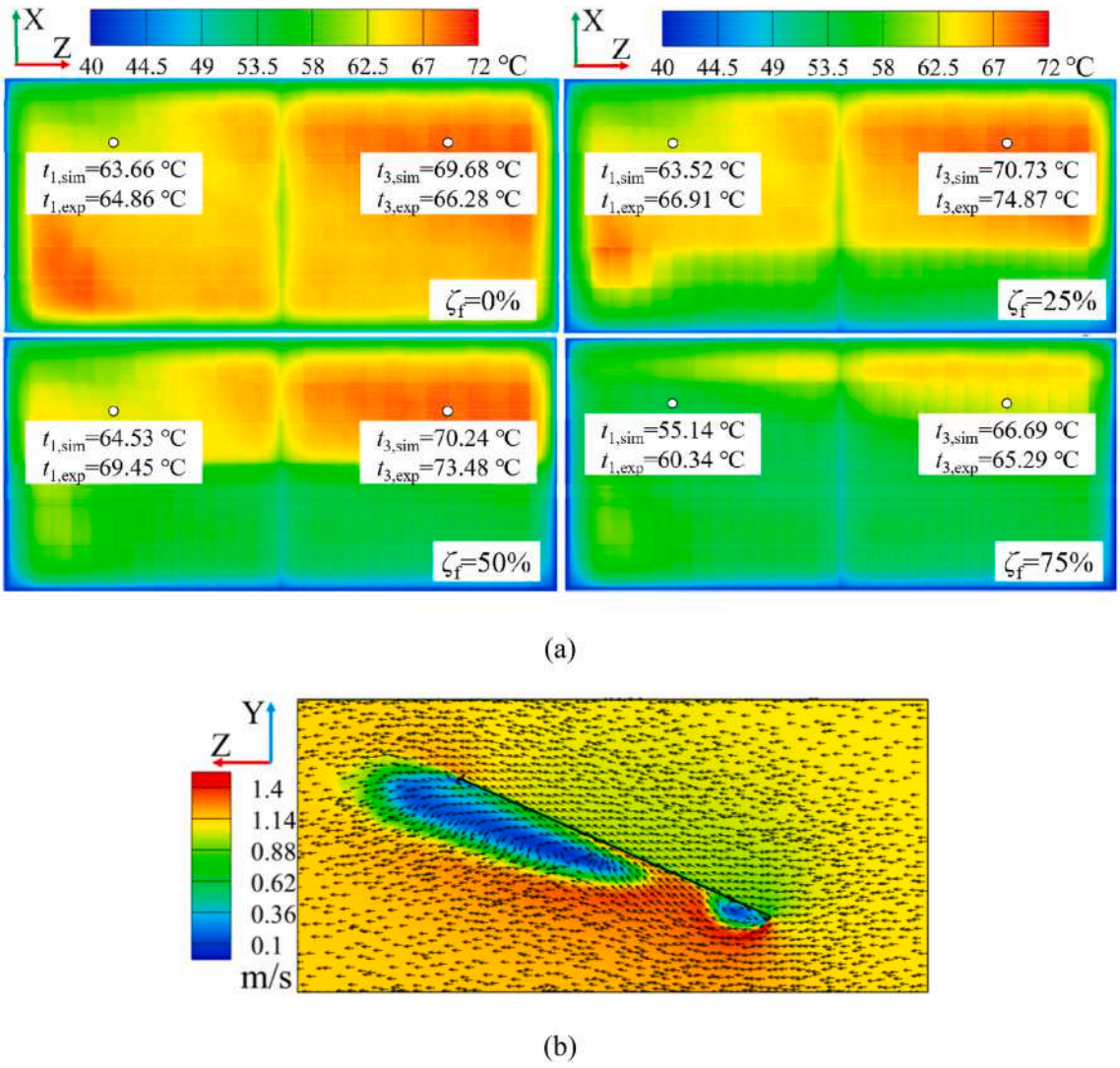


Fig. 20. Temperature and flow fields characteristics at 13:00 under different vertical shading ratios: (a) Temperature distribution on BHCPV modules; (b) Flow field around BHCPV module.

irradiance and high-temperature environments, leading to deviations in optical measurements during dynamic testing. Secondly, the actual light field is affected by the combined influence of variable sky conditions and complex surrounding environments, especially the multiple reflections and scattering effects on rear side, which are difficult to capture accurately in optical simulations, thus introducing inherent uncertainties.

Electrical errors: As a secondary source of systematic errors, the impact of electrical errors is equally significant. On one hand, the measurement results of I-V curve tester are affected by NUT and NUIR on BHCPV module. On the other hand, under partial shading conditions, the nonlinear response and reconstruction behavior of the internal circuits are extremely complex. This places higher demands on the accurate solution of IM-DEC model, increasing the simulation difficulty and potential errors.

Thermal errors: The thermal model error is relatively minor, but not negligible. Besides the inherent hysteresis of temperature sensor, simplifying the three-dimensional dynamic wind field to a uniform planar wind field in simulation may lead to inaccuracies in the estimation of local convective heat transfer coefficients, thus causing temperature prediction errors at some measurement points.

Therefore, future research will focus on improving the accuracy of optical simulations, such as by considering more complex sky models and backlight fields, and further improving the accuracy of iterative calculations of the optical-thermal-electrical-environmental coupling process.

5.6. *A summary of comparisons with existing models*

The existing models exhibit several limitations. In terms of PV types, most studies focus on MFCPV, MHCPV, and BFCPV modules, with insufficient attention given to BHCPV modules. In optical models, irradiance distributions are often treated as uniform or overly simplified. Thermal models typically assume unidirectional coupling from environment to system, lack a true environmental domain, and cannot obtain a realistic temperature distribution. Electrical models frequently simplify or ignore the impacts of non-uniformities. Thermal-electrical coupling is usually unidirectional, applied at the module level, and lacks dynamic feedback.

The proposed model in this study offers significant advantages. This model is designed for BHCPV modules and can also be applied to other types of PV modules, taking into account the effects of front-side shading and rear-side NUIR. To further demonstrate the applicability of IM-DEC model to different types of PV modules, based on the circuit topology in Ref. [16], the electrical characteristics of MFCPV module under different shading conditions are investigated using IM-DEC model, and compared with the experimental results. MFCPV module model is CNPV-240M,

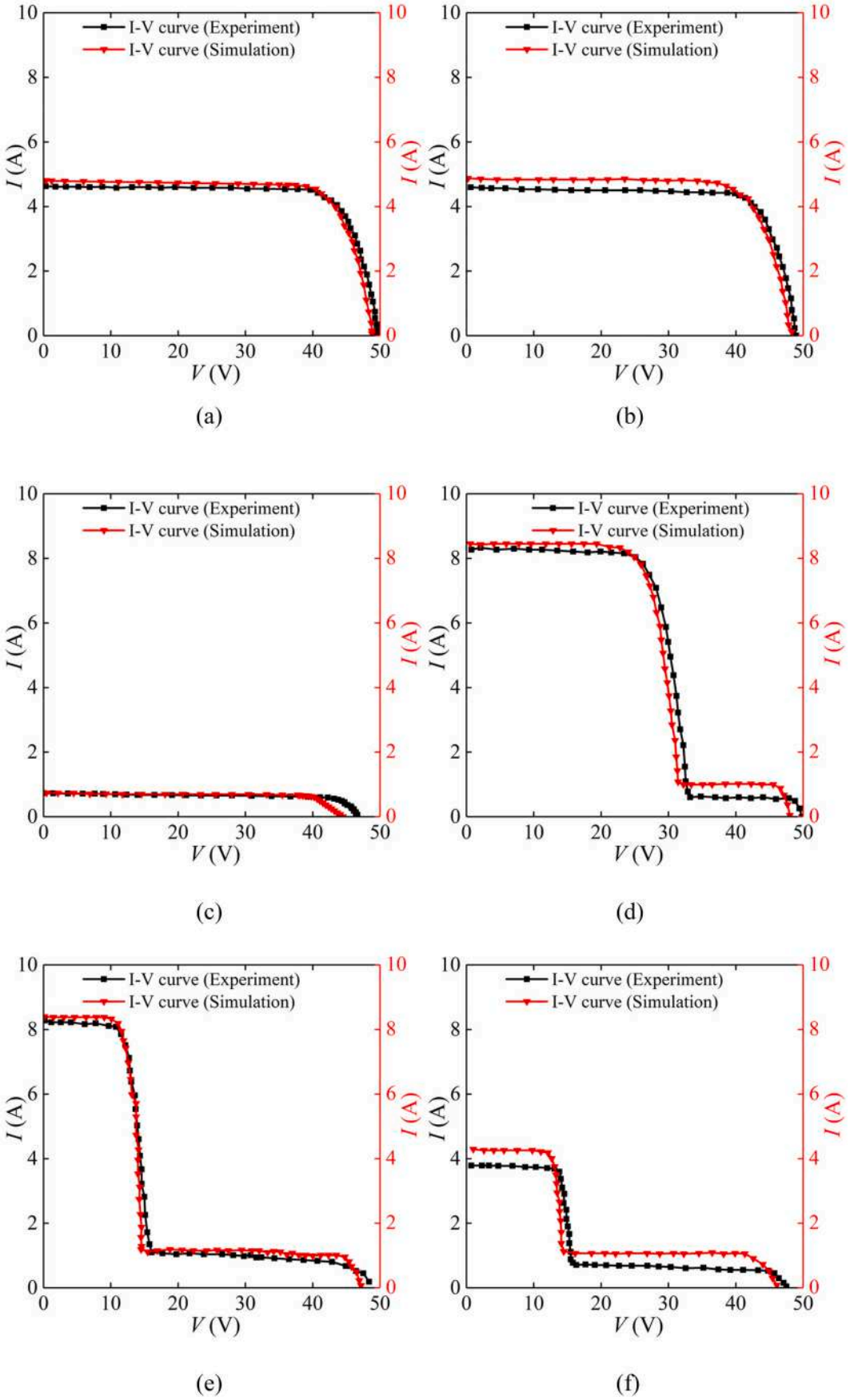


Fig. 21. Comparison of I-V curves under different shading conditions on November 20, 2025 between simulation and experiment: (a) Horizontal shading ($\zeta_f = 25\%$); (b) Horizontal shading ($\zeta_f = 50\%$); (c) Horizontal shading ($\zeta_f = 75\%$); (d) Vertical shading ($\zeta_f = 25\%$); (e) Vertical shading ($\zeta_f = 50\%$); (f) Vertical shading ($\zeta_f = 75\%$).

and its circuit topology is the most conventional series arrangement. It has 60 FCEs, divided into 3 bypass diode protection strings (each string containing 20 series-connected FCEs). The shading patterns are divided into vertical shading and horizontal shadings, with the shading ratio increasing from 0% to 100%. The simulation conditions are consistent with the experimental conditions in Ref. [16], and the irradiance is converted to the standard reference condition of 1000 W/m^2.

Fig. 22(a) and (b) show the P of MFCPV module calculated using IM-DEC model under different vertical and horizontal shading ratios, respectively, and these results are compared with the experimental findings from Ref. [16]. It is shown that the maximum relative error between simulation and experiment under vertical and horizontal shadings is 3.36% and 3.35%, respectively. Additionally, Fig. 22(c) and (d) present the I-V curves of MFCPV module when a single FCE is shaded by 50% and 100%, respectively, alongside curves from Ref. [16]. It can be seen that, due to the influence of shading, the I-V curve exhibits characteristic knees, which matches the actual electrical characteristics. The RMSE of the current in I-V curve is 0.47 A and 0.34 A, respectively. In summary, the small relative error proves the correctness and generalizability of IM-DEC model.

6. Conclusions

An IM-DEC model was proposed for BHCPV modules to predict electrical performance under partial shading conditions. The model accounted for the non-uniformity of irradiance and temperature among HCEs, introduced a dynamic thermal-electrical coupling. It has been experimentally validated under different shading ratios and patterns. Furthermore, the model was compared with the existing CO-DEC and AVT models. The conclusions are as follows:

Under horizontal shading: for shading ratios (ζ_f) of 0%, 25%, 50% and 75%, the average relative errors of AVT model are 6.04%, 29.21%, 6.06% and 107.15%, respectively; the average relative errors of CO-DEC model are 3.72%, 11.18%, 15.26% and 23.99%, respectively; the average relative errors of IM-DEC model are 2.93%, 4.41%, 5.00%, and 6.35%, and the overall accuracy is the best.

Under vertical shading: for ζ_f of 0%, 25%, 50% and 75%, the average relative errors of AVT model are 4.50%, 26.18%, 91.49% and 90.81% respectively; the average relative errors of CO-DEC model are 8.29%, 11.17%, 12.33% and 25.72%; the average relative errors of IM-DEC model are 4.69%, 5.86%, 6.43%, and 6.42%, and the accuracy is significantly higher than the previous two.

The maximum relative errors between $G_{GHI,sim}$ and $G_{GHI,exp}$ obtained through optical-thermal-electrical-environmental multiphysics

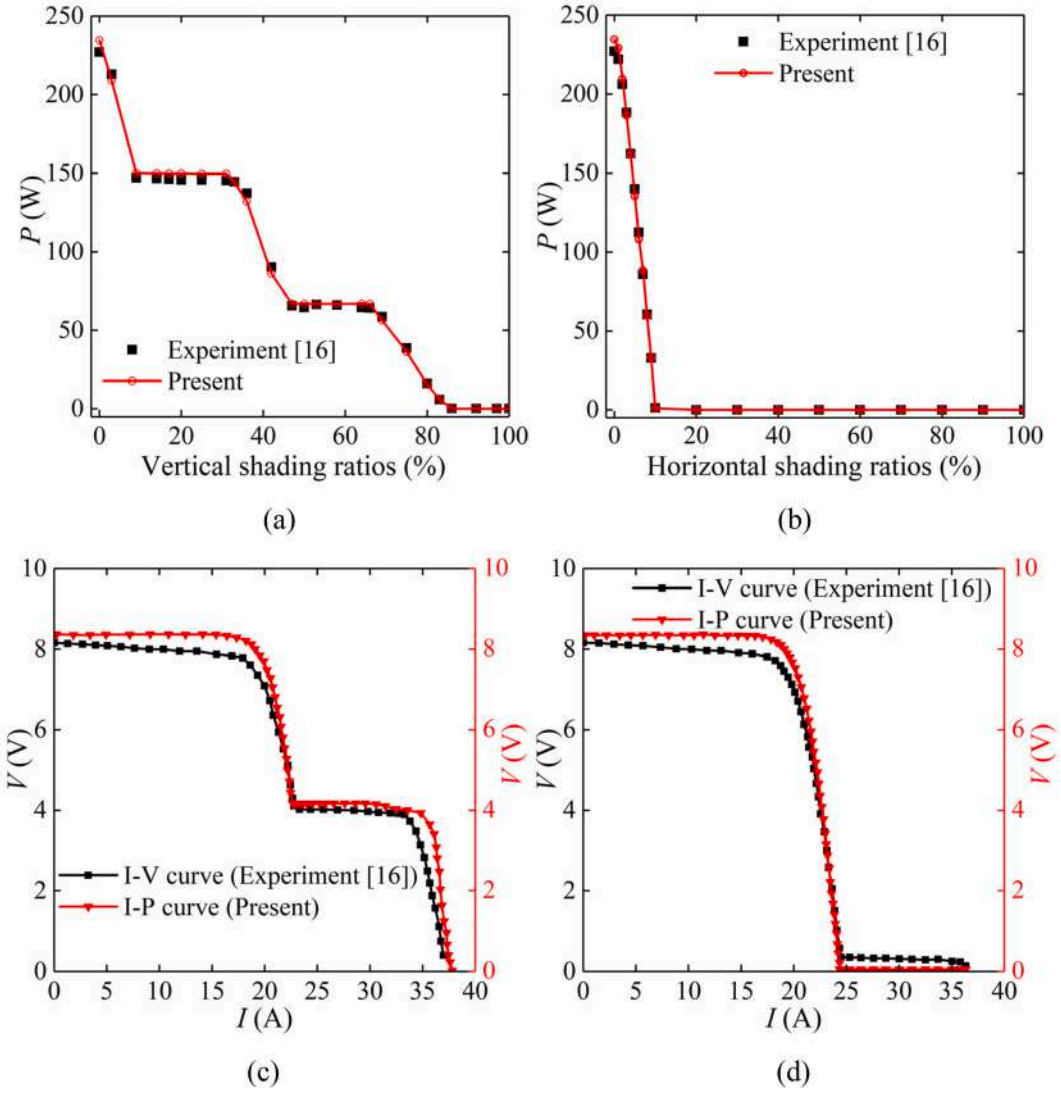


Fig. 22. Comparison of electrical characteristics for MFCPV module under different shading conditions between IM-DEC model simulation and experiment [16]: (a) Vertical shading; (b) Horizontal shading; (c) 50% shading by a single FCE; (d) 100% shading by a single FCE.

calculations based on IM-DEC model are all less than 11.7% under both horizontal and vertical shading, demonstrating the validity of the optical model. The maximum relative errors between the calculated and measured temperatures at monitoring points are all below 10.6%, thus proving the correctness of the thermal performance calculation.

AVT model is based on the principle of energy conversion and assumes that all strings operate under normal conditions. It cannot reflect the electrical non-uniformity within strings and dynamic activation of bypass diodes. Moreover, the circuit reconstruction caused by local mismatch is ignored. Since CO-DEC model uses average irradiance and temperature as inputs, it results in significant inaccuracies under complex shading conditions. Due to inherent limitations in the irradiance and temperature input mechanisms, AVT and CO-DEC models are unable to accurately reflect the circuit reconstruction and nonlinear I-V features caused by partial shading, which are more suitable for electrical performance prediction under non-shading conditions.

IM-DEC model in this study takes into account the NUIR and NUT distributions based on half-cell element, and accurately characterizing the impact of local mismatch on electrical performance through a dynamic thermal-electrical coupling. It can capture the activation of bypass diodes and the associated multi-knees I-V characteristics. The RMSE of current in the I-V curve are less than 0.56 A, for different operating conditions. Furthermore, the accuracy and universality of this model are further demonstrated by verifying it against the circuit structure in literature. This approach provides theoretical support and technical guidance for the performance assessment of BHCPV modules.

Finally, it should be pointed out that the IM-DEC model proposed in this study has only been validated for the performance prediction of individual module under horizontal and vertical shadings. However, the system-level (such as full PV arrays) generalization of this model under complex shading conditions such as diagonal, random, time-varying shadings and shadow-based occlusion scenarios still needs further verification, which will increase the solution difficulty and parallel computing cost. Also, this model can be further considered to be effectively integrated into digital design and optimization processes to enhance its practical application value and scalability.

CRediT authorship contribution statement

Shi-Jie Xu: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Shuang-Ying Wu:** Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Lan Xiao:** Writing – review & editing, Resources, Funding acquisition. **Yan Hu:** Software.

Declaration of competing interest

■ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

Abbreviations

AVT: Average temperature
BFCPV: Bifacial full-cell photovoltaic
BHCPV: Bifacial half-cell photovoltaic
BPV: Bifacial photovoltaic
CO-DEC: Conventional discrete equivalent circuit
EC: Equivalent circuit
EVA: Ethylvinylacetate
FCE: Full-cell element
FVM: Finite volume method
HCE: Half-cell element
IM-DEC: Improved discrete equivalent circuit
KCL: Kirchhoff's current law
KVL: Kirchhoff's voltage law
LCOE: Levelized cost of solar energy
MFCPV: Monofacial full-cell photovoltaic
MHCPV: Monofacial half-cell photovoltaic
MPV: Monofacial photovoltaic
NUIR: Non-uniform irradiance
NUT: Non-uniform temperature
PV: Photovoltaic
RMSE: Root mean square error
SC: Silicon cell
SPICE: Simulation program with integrated circuit emphasis
STC: Standard test conditions
UDF: User-defined function

Symbols

d: Thickness of BHCPV module (mm)
E_g: Material energy bandgap (eV)
$E_{g,ref}$: Material energy bandgap at reference temperature (eV)
G: Irradiance (W/m^2)
G_{DHI}: Diffuse horizontal irradiance (W/m^2)
G_{DNI}: Direct normal irradiance (W/m^2)
G_e: Equivalent irradiance (W/m^2)
G_{GHI}: Global horizontal irradiance (W/m^2)
G_{ref}: Solar irradiance of 1000 W/m^2 under STC
h_{mgd}: Minimum ground distance (mm)
I: Current (A)
I_d: Currents flowing through diode (A)
I_{HCE}: Currents of HCE (A)
I_0: Diode reverse saturation current (A)
$I_{0,ref}$: Diode reverse saturation current under STC (A)
I_{ph}: Photocurrent (A)
$I_{ph,ref}$: Photocurrent under STC (A)
$I_{sc,ref}$: Short circuit current under STC (A)
I_{sh}: Currents flowing through shunt resistor (A)
I_{total}: The current of BHCPV module (A)
j: Number of iterations
k: Boltzmann constant
l: Length of BHCPV module (mm)
m: Number of HCEs in BHCPV module
P: The maximum instantaneous electric power of BHCPV module (W)
P_i: Instantaneous electric power of i-th HCE (W)
q: Electron charge (C)
q_i: Energy converted into heat energy in i-th HCE (W/m^2)
q_{SC}: Energy converted into heat energy in SC layer (W/m^2)
R_s: Series resistor (Ω)
$R_{s,ref}$: Series resistance under STC (Ω)
R_{sh}: Shunt resistor (Ω)
$R_{sh,ref}$: Shunt resistor under STC (Ω)
S: Shading areas (m^2)
t: Temperature (°C)
t_{amb}: Ambient temperature (°C)
t_{ref}: Reference temperature (°C)
V: Voltage (V)
V_D: The turn-on threshold of the bypass diode (V)
V_{HCE}: Voltage of HCE (V)
V_{total}: The voltage of BHCPV module (V)
$V_{oc,ref}$: Open circuit voltage under STC (V)
w: Width of BHCPV module (mm)

Greeks

α: Wind direction angle (°)
α_{Isc}: Short circuit current temperature coefficient
α_{Rs}: Temperature coefficient of internal series resistance
θ: Tilt angle (°)
β: Absorptivity
τ: Transmissivity
η_{el}: Instantaneous electrical efficiency (%)
η_{ref}: Photovoltaic conversion efficiency at reference temperature (%)
δ: Temperature coefficient (%)
φ: Bifaciality factor
ζ: Shading ratios (%)
Δl: HCE length (mm)
Δw: HCE width (mm)

Subscripts

ave: Average
BFCPV: BFCPV module
exp: Experimental results
f: Front side
g: Glass cover
i: i-th HCE
r: Rear side
Sim: Simulation results